Foundations of Large-Scale Multimedia Information Management and Retrieval

Lecture #6 Scalability

Edward Y. Chang

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Foundations of Large-Scale Multimedia Information Management and Retrieval
Mathematics of Perception

Foundations of Large-Scale Multimedia Information Management and Retrieval *Mathematics of Perception* covers knowledge representation and semantic analysis of multimedia data and scalability in signal extraction, data mining, and indexing. The book is divided into two parts: Part I - Knowledge Representation and Semantic Analysis focuses on the key components of mathematics of perception as it applies to data management and retrieval. These include feature selection/reduction, knowledge representation, semantic analysis, distance function formulation for measuring similarity, and multimodal fusion. Part II - Scalability Issues presents indexing and distributed methods for scaling up these components for high-dimensional data and Web-scale datasets. The book presents some real-world applications and remarks on future research and development directions.

The book is designed for researchers, graduate students, and practitioners in the fields of Computer Vision, Machine Learning, Large-scale Data Mining, Database, and Multimedia Information Retrieval.

Dr. Edward Y. Chang was a professor at the Department of Electrical & Computer Engineering, University of California at Santa Barbara, before he joined Google as a research director in 2006. Dr. Edward Y. Chang received his M.S. degree in Computer Science and Ph.D degree in Electrical Engineering, both from Stanford University.

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Springer

More Data vs. Better Algorithms

Banko & Brill, 2001

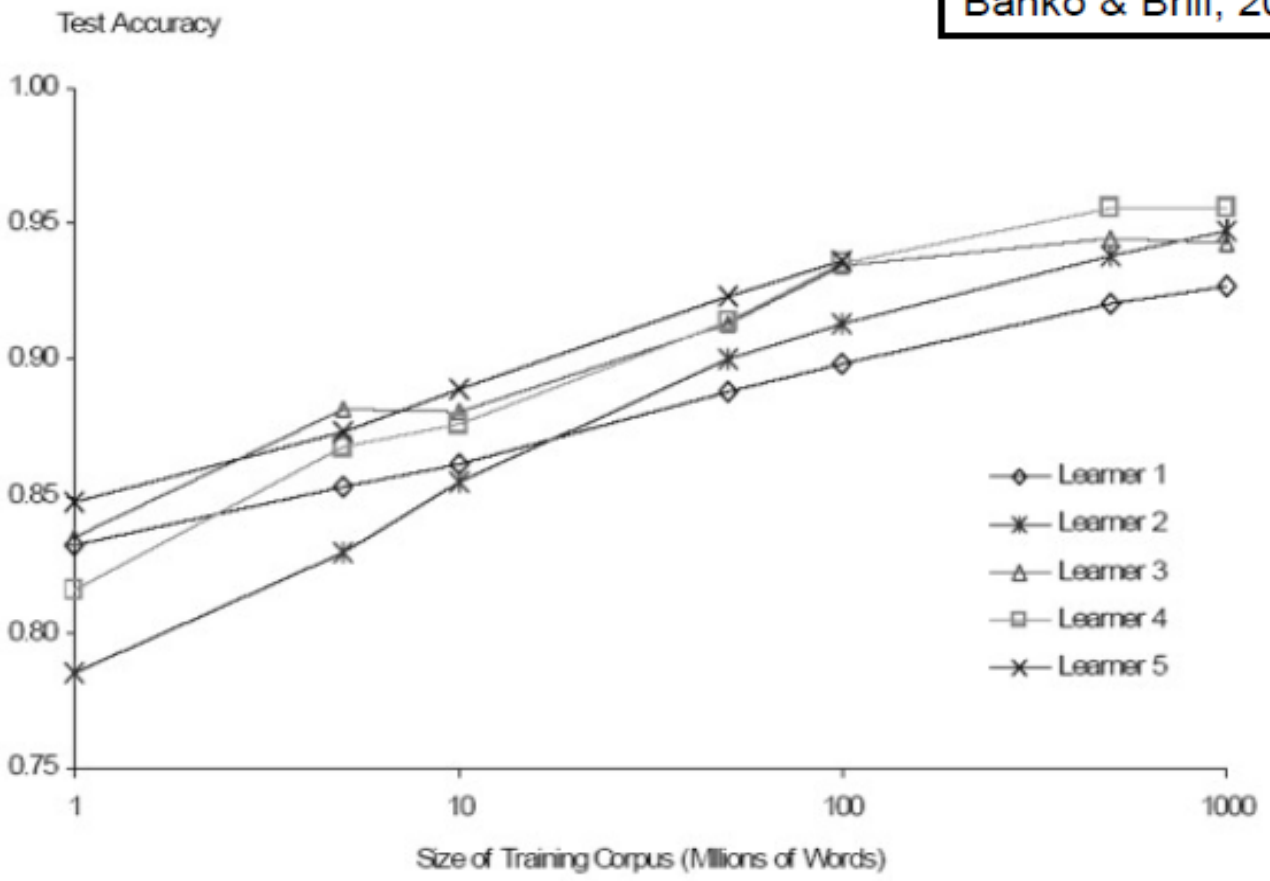


Figure 2. Learning Curves for Confusable Disambiguation

More Data vs. Better Algorithms

Banko & Brill, 2001

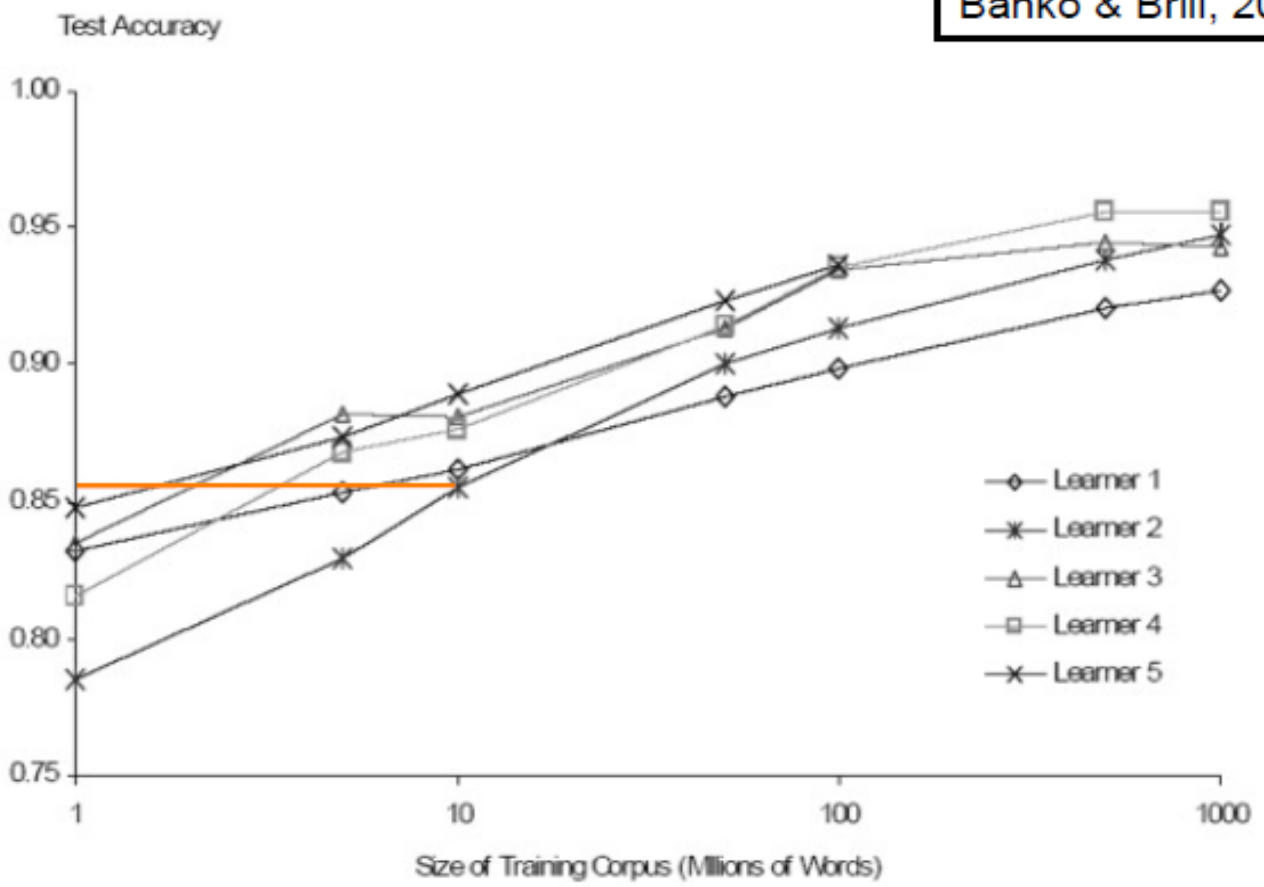


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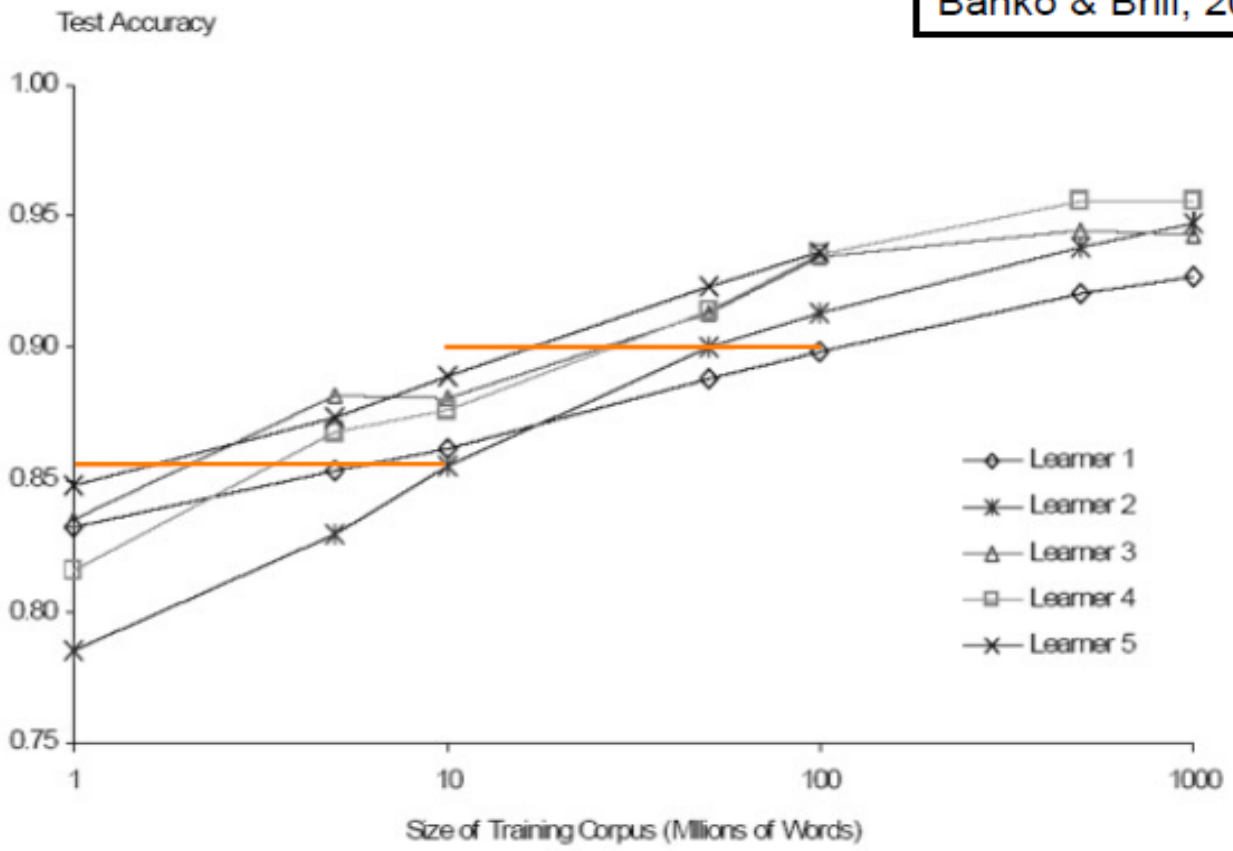


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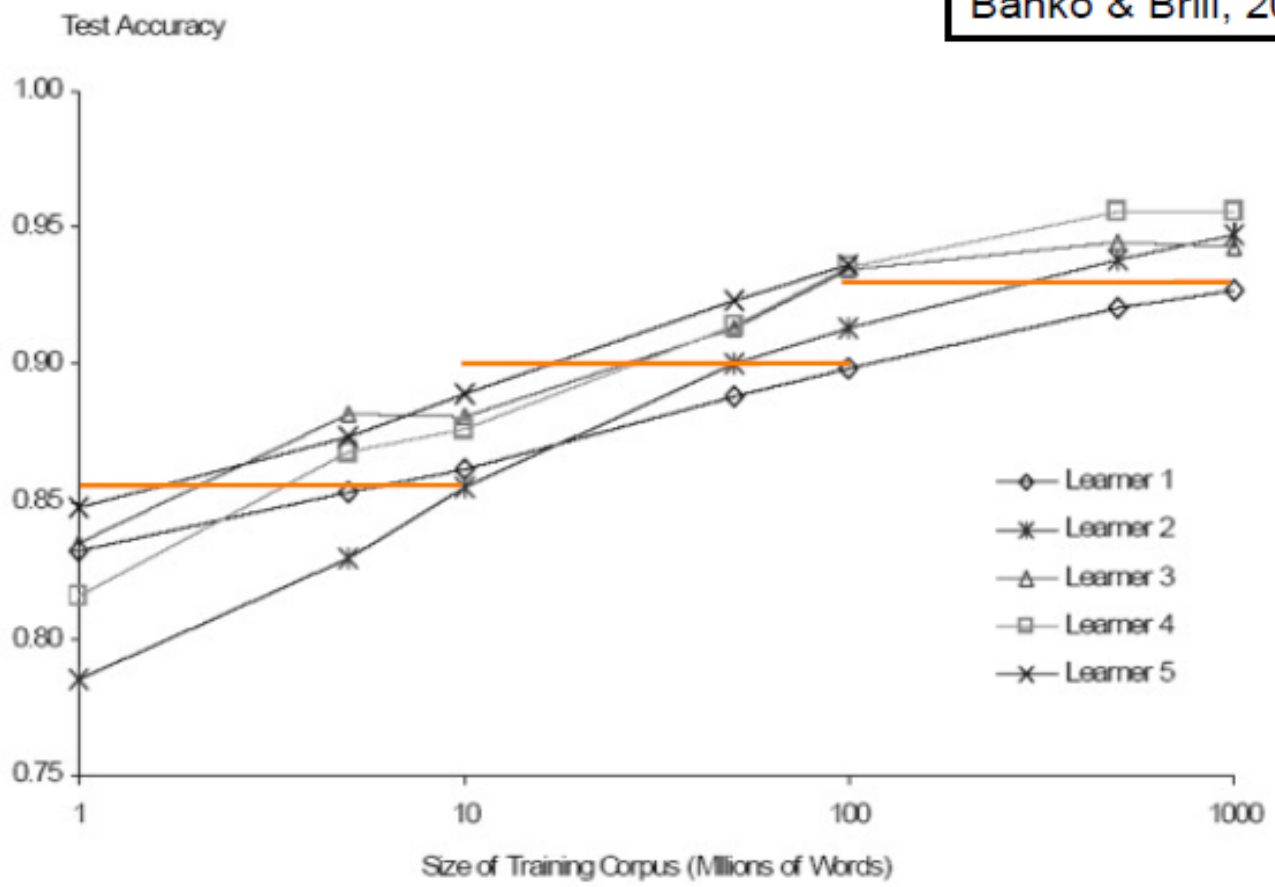
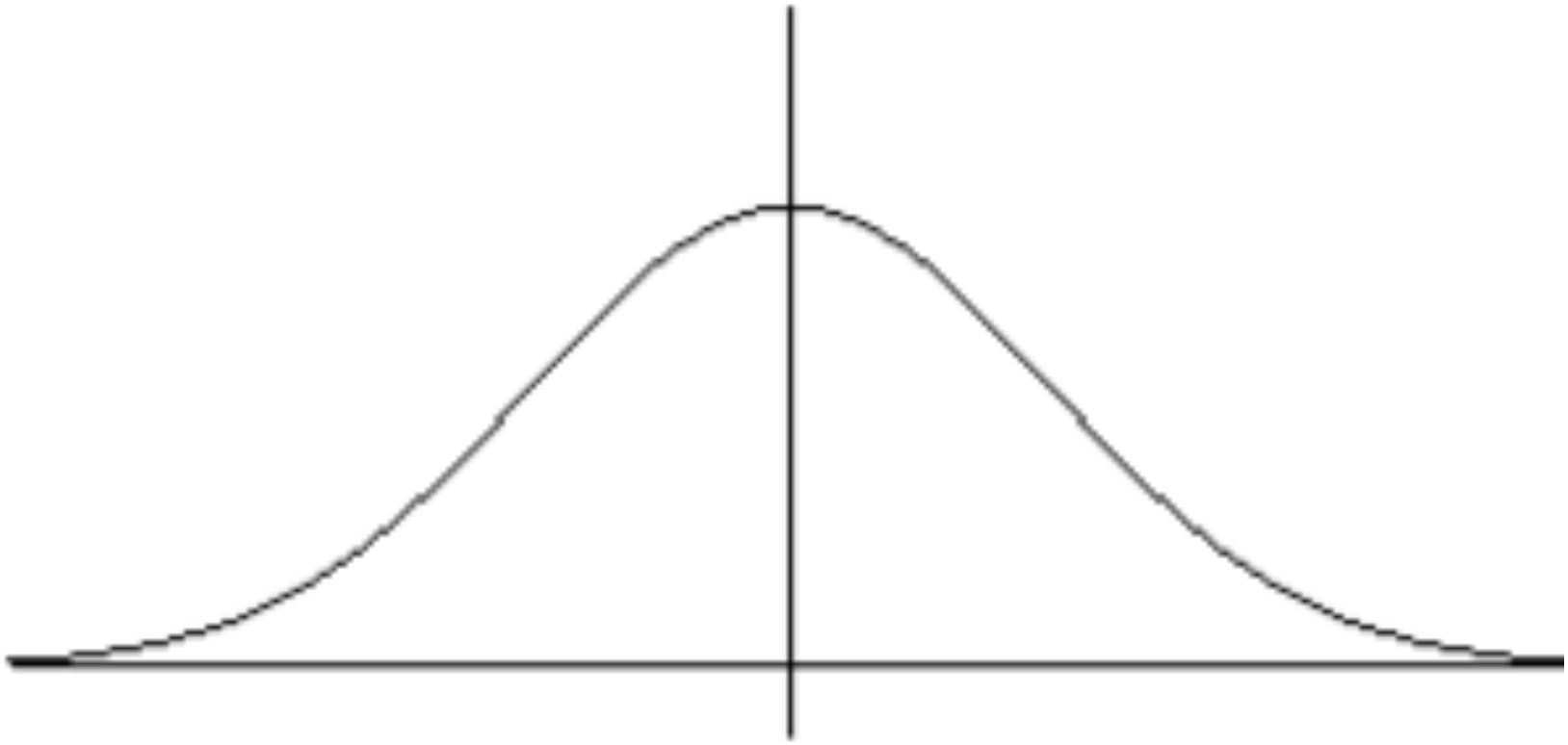


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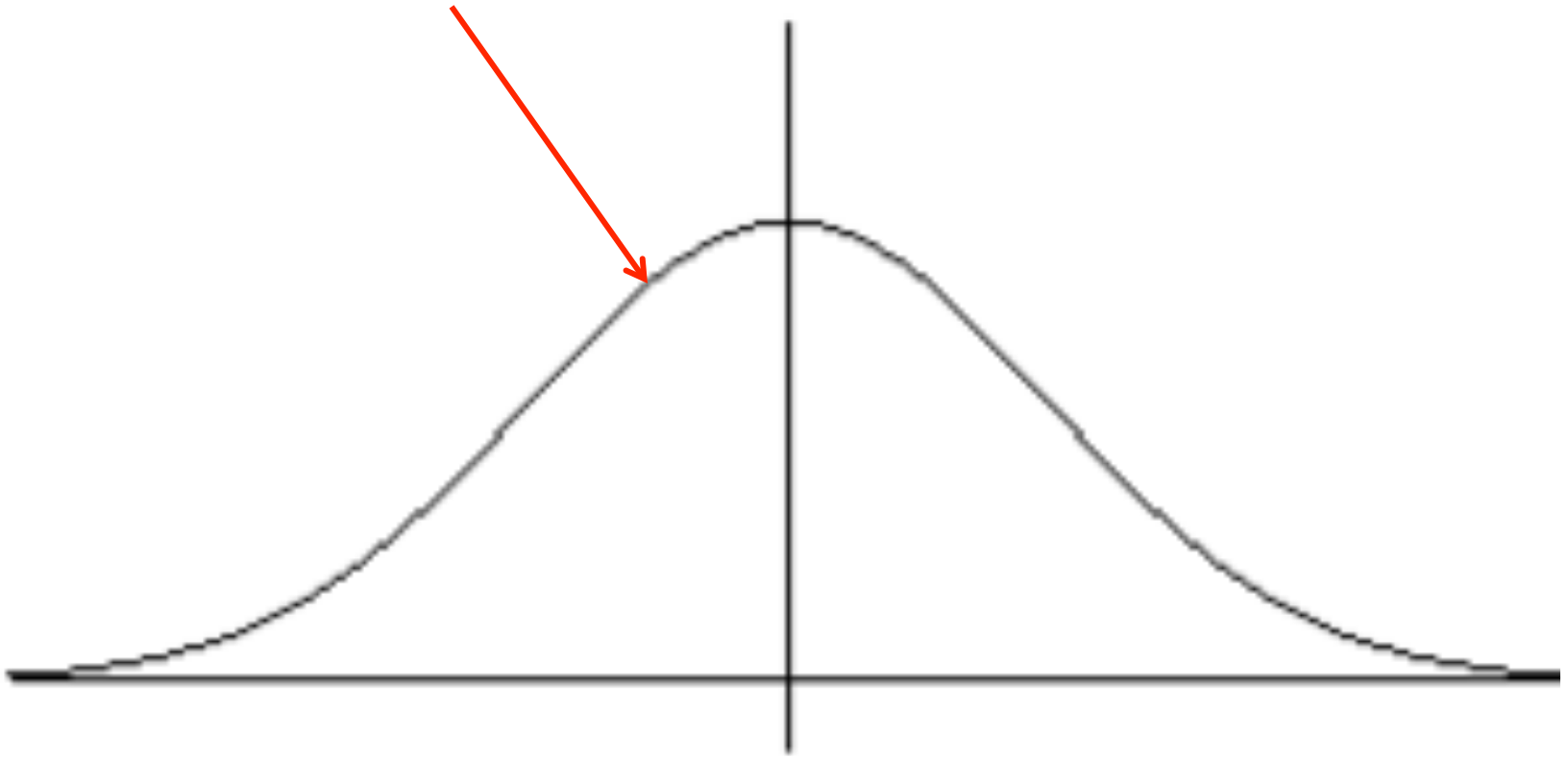
How Much Data?

Jerry Friedman



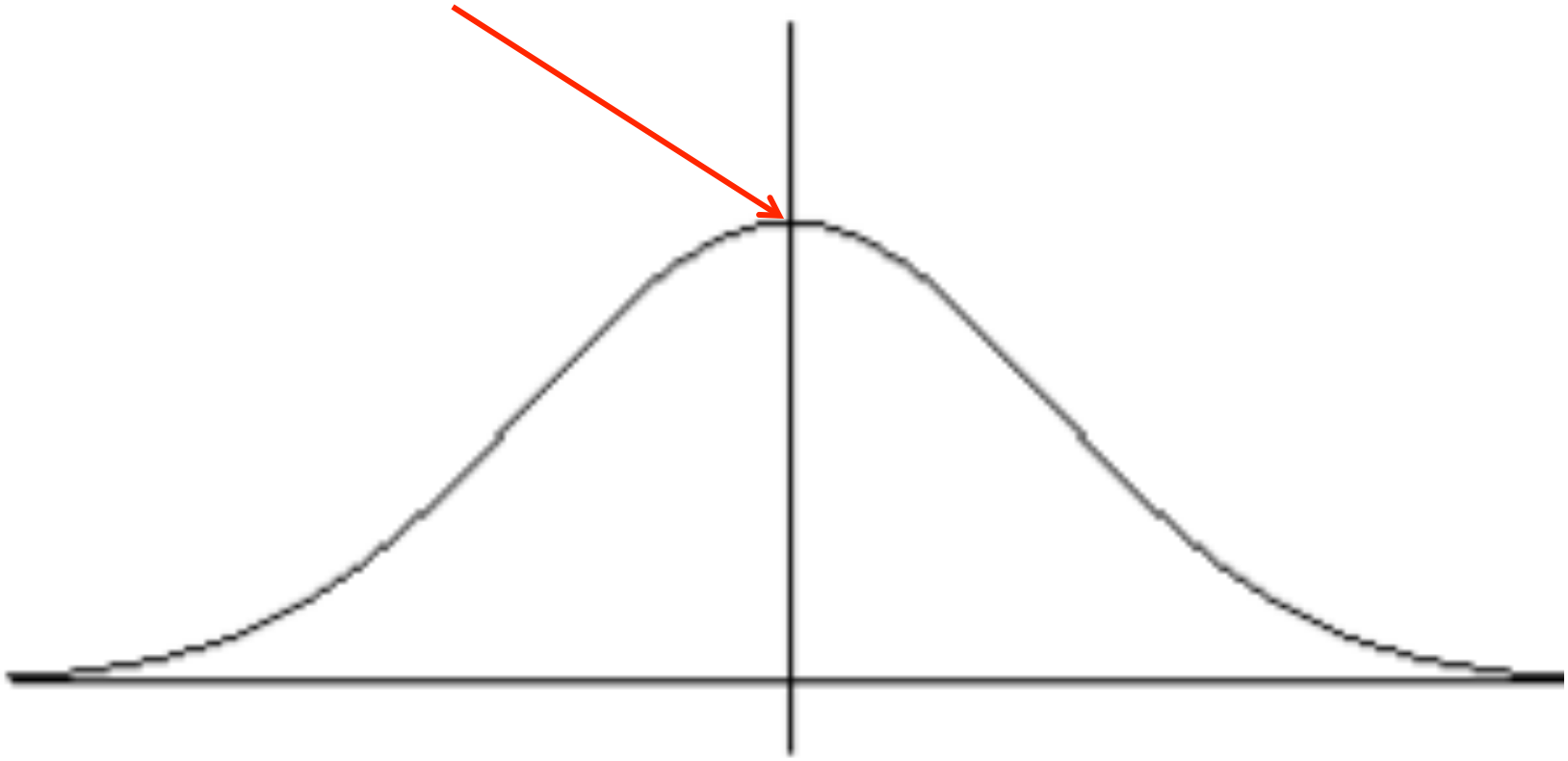
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How Much Data?

Jerry Friedman



Outline

- Motivating Applications
  - Image Annotation
  - Social Media
- Key Subroutines
  - Frequent Itemset Mining [ACM RS 08]
  - Latent Dirichlet Allocation [WWW 09, TIST 10]
  - Support Vector Machines [NIPS 07]
  - Clustering [ECML 08, PAMI 10]
  - UserRank [WWW 10, VLDB 10]
- Distributed Computing Perspectives

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Related Publications

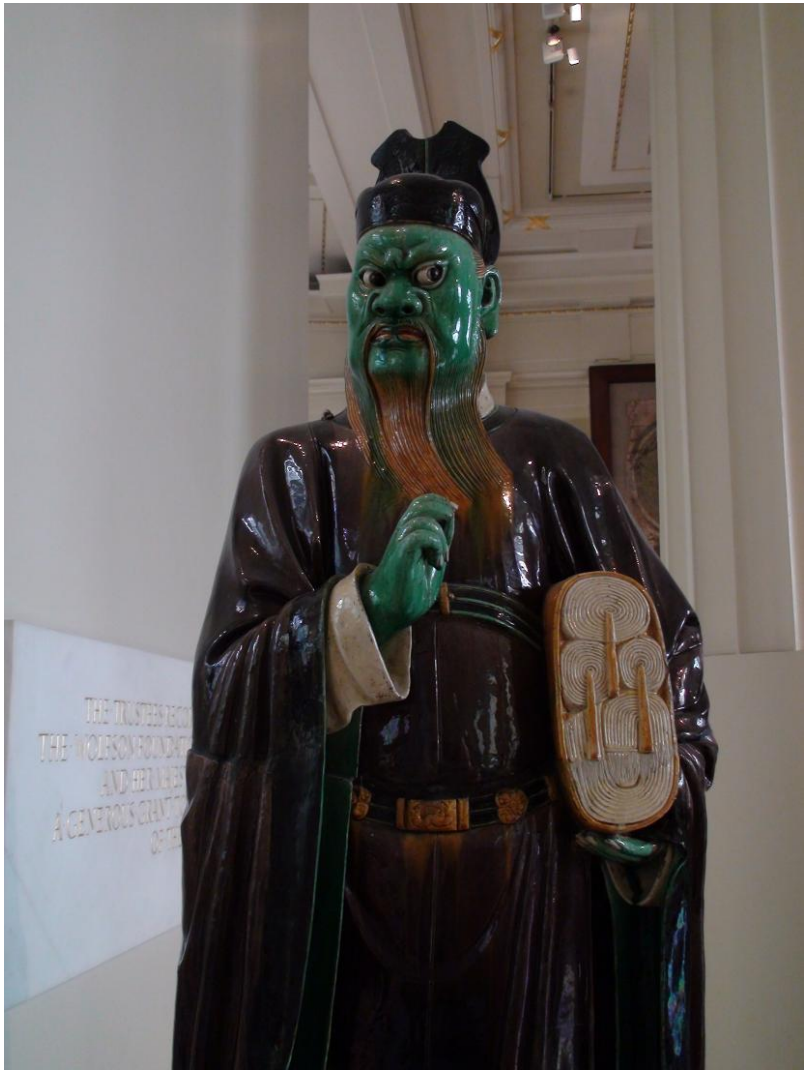
- **Large-scale Collaborative Filtering**:
  - PLDA+: Parallel Latent Dirichlet Allocation for Large-Scale Applications, ACM Transactions on Intelligent Systems and Technology (TIST), 2010.
  - Collaborative Filtering for Orkut Communities: Discovery of User Latent Behavior, W.-Y. Chen, J. Chu, Y. Wang, and E. Y. Chang, WWW 2009: 681-690.
  - PFP: Parallel FP-Growth for Query Recommendation, H. Li, Y. Wang, D. Zhang, M. Zhang, and E. Y. Chang, ACM Recommendation Systems, Lausanne, October 2008
  - PVSM --- Parallelizing SVMs on Distributed Machines, E. Y. Chang, et al., NIPS 2007.
  - Parallel Spectral Clustering in Distributed Systems,
    Wen-Yen Chen, Yangqiu Song, Hongjie Bai, Chih-Jen Lin, and E. Y. Chang,
    IEEE Transactions on Pattern Analysis and Machine Intelligence (PAMI), 2010.
- **Social Media**:
  - A Deep Model-based and Data-driven Hybrid Architecture for Image Annotation,
    Zhiyu Wang, Dinying Xia, E. Y. Chang, International Workshop on Very-Large-Scale Multimedia Corpus, Mining and Retrieval, with ACM Multimedia, Florence, October 2010.
  - AdHeat: An Influence-based Diffusion Model for Propagating Hints to Match Ads,
    H.J. Bao and E. Y. Chang, WWW 2010 (best paper candidate), April 2010.
  - Confucius and its Intelligent Disciples, Integrating Search and Social, X. Si, E. Y. Chang, Z. Gyongyi, and M.S. Sun, VLDB, September 2010 .
  - Combinational Collaborative Filtering for Personalized Community Recommendation, W.-Y. Chen, D. Zhang, and E. Y. Chang, KDD 2008: 115-123.



User Experience Study Reveals…

- Keyword-based
  - Preferred query method
  - Poor refinement method
- Content-based
  - Preferred refinement method
  - Poor query method
- But…
  - Most media data are w/o keywords
  - Keywords can be semantically ambiguous





Challenges

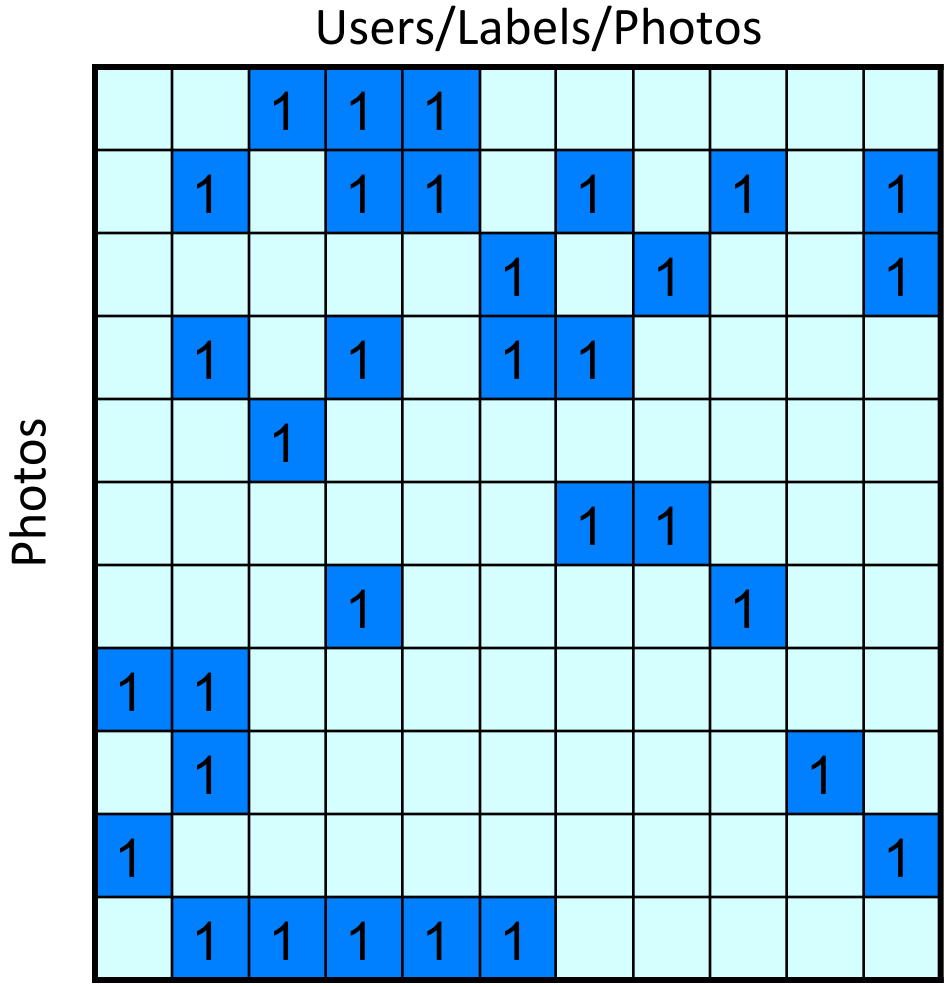
- Feature Extraction
- Similarity or Distance Function
- Training Algorithms
  - Model based
  - Data driven
- Training Data
  - How much
- Scalability

Collaborative Filtering

Based on *membership* so far,
and *memberships* of others

↓

Predict further *membership*



Collaborative Filtering

Based on *partially* observed matrix

↓

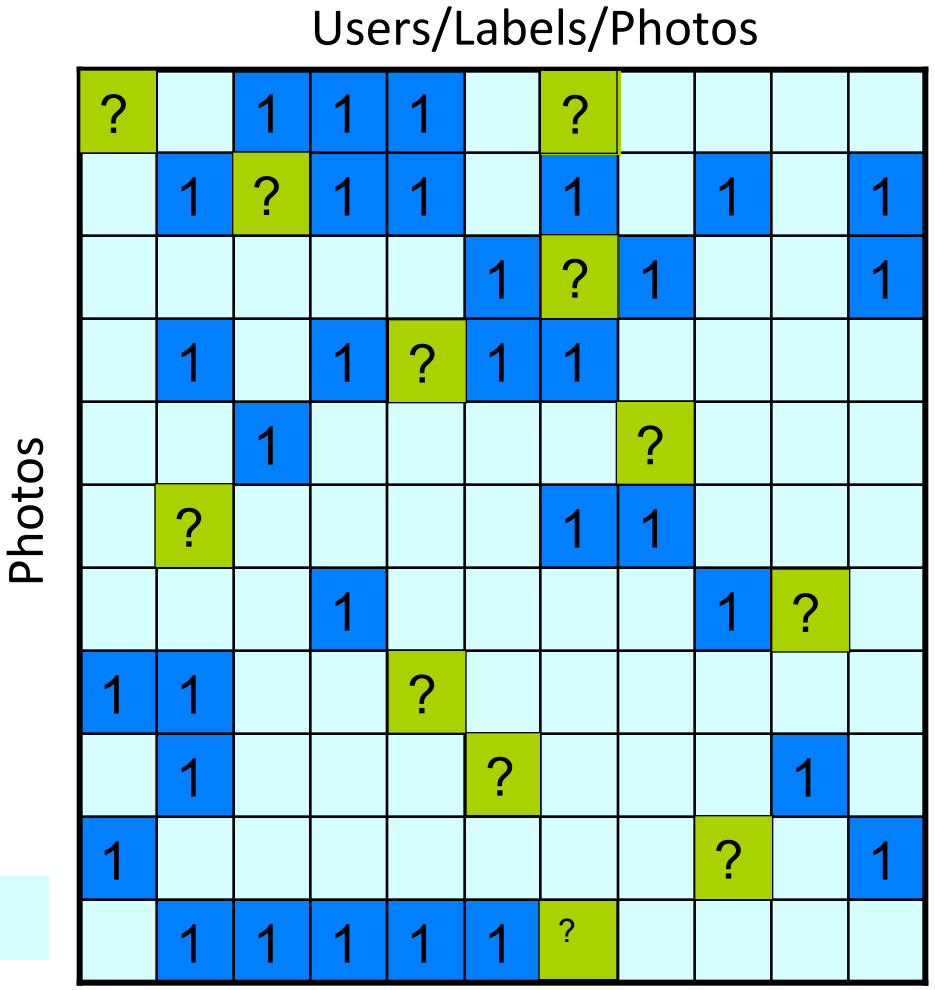
Predict *unobserved* entries

↓

I. Will user **i** enjoy photo **j**?

II. Will user **i** be interesting to user **j**?

III. Will photo **i** be related to photo **j**?



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FIM-based Recommendation







To grow the base, we need association rules

- An association rule: $a, b, c \longrightarrow d$
- A Bayesian interpretation: $P(d \mid a, b, c) = \frac{N(a,b,c,d)}{N(a,b,c)}$
- The key is to count the occurrences (*support*) of itemsets $N(\ldots)$

FIM Preliminaries

- Observation 1: If an item A is not frequent, any pattern contains *A* won't be frequent [R. Agrawal]

  ➔ use a threshold to eliminate infrequent items

  {*A*} → {*A, B*}
- Observation 2: Patterns containing *A* are subsets of (or found from) transactions containing *A* [J. Han]

  ➔ divide-and-conquer: select transactions containing A to form a conditional database (CDB), and find patterns containing A from that conditional database

  {*A, B*}, {*A, C*}, {*A*} → CDB *A*

  {*A, B*}, {*B, C*} → CDB *B*

Preprocessing

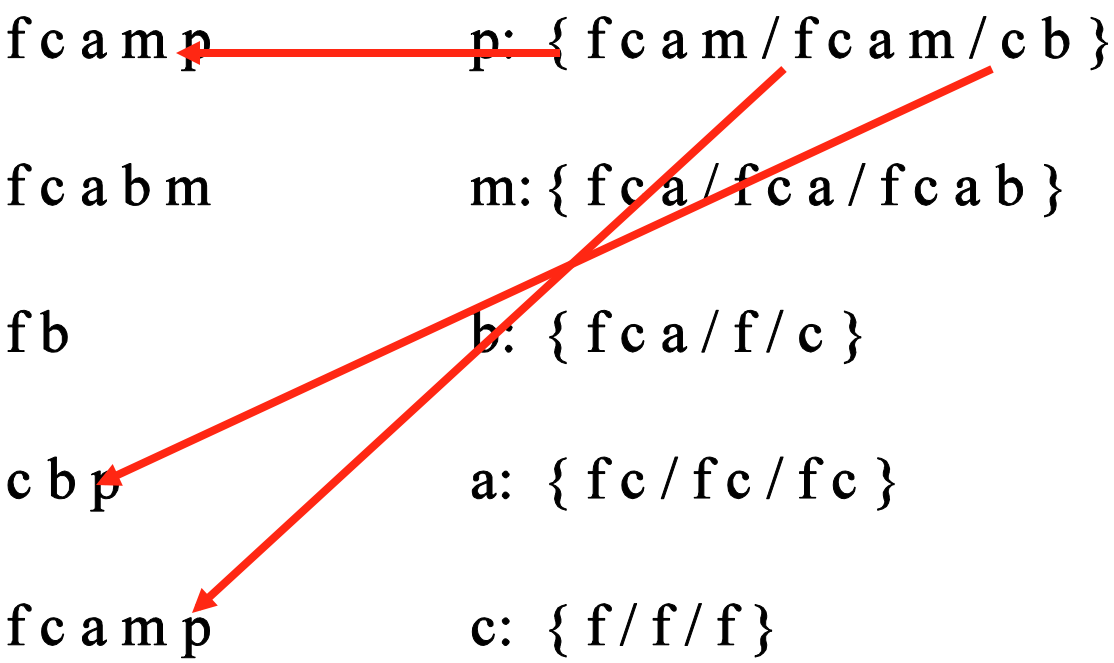
	f: 4 c: 4 a: 3 b: 3 m: 3 p: 3	
f a c d g i m p		f c a m p
a b c f l m o		f c a b m
b f h j o	o: 2 d: 1 e: 1 g: 1 h: 1 i: 1 k: 1 l : 1 n: 1	f b
b c k s p		c b p
a f c e l p m n		f c a m p

- According to Observation #1, we count the support of each item by scanning the database, and eliminate those infrequent items from the transactions.
- According to Observation #3, we sort items in each transaction by the order of descending support value.

Parallel Projection

- According to Observation 2, we construct CDB of item A; then from this CDB, we find those patterns containing *A*
- How to construct the CDB of *A*?
  - If a transaction contains *A*, this transaction should appear in the CDB of *A*
  - Given a transaction {*B, A, C*}, it should appear in the CDB of *A*, the CDB of *B*, and the CDB of *C*
- Dedup solution: using the order of items:
  - sort {*B,A,C*} by the order of items → <*A,B,C*>
  - Put <> into the CDB of *A*
  - Put <*A*> into the CDB of *B*
  - Put <*A,B*> into the CDB of *C*

Example of Projection

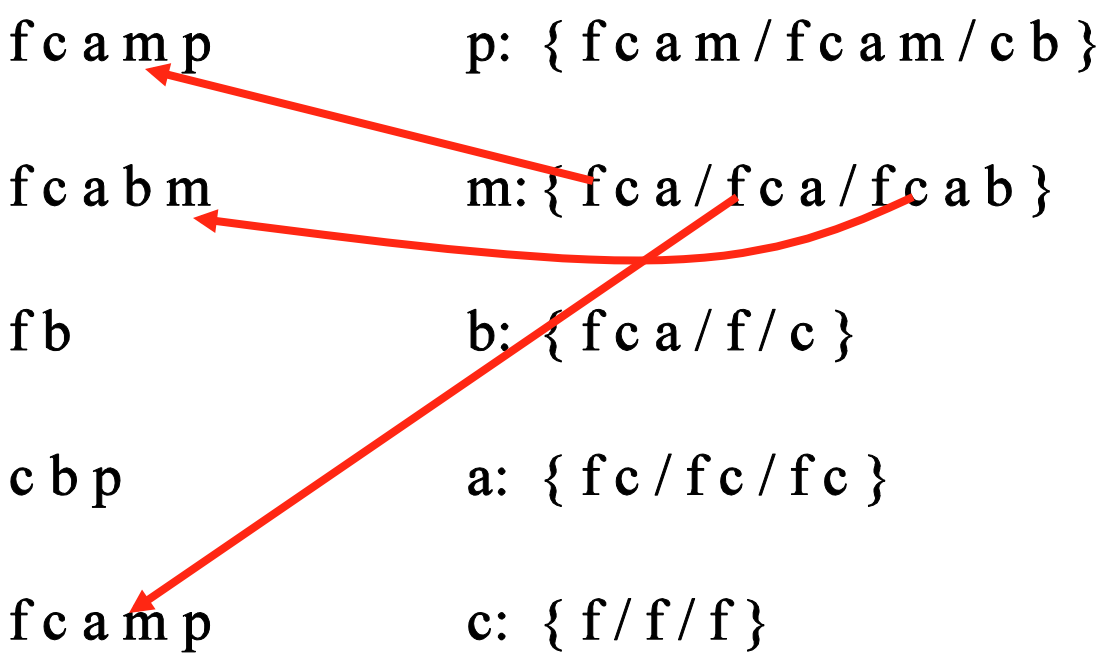


Example of Projection of a database into CDBs

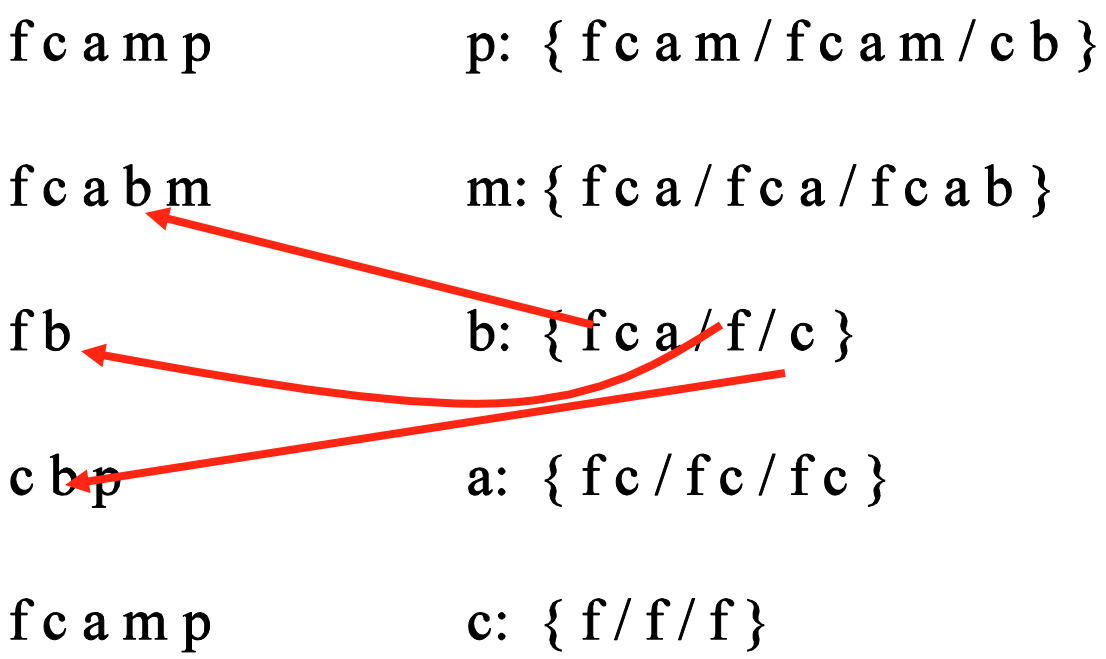
Left: sorted transactions in order of *f*, *c*, *a*, *b*, *m*, *p*

Right: conditional databases of frequent items

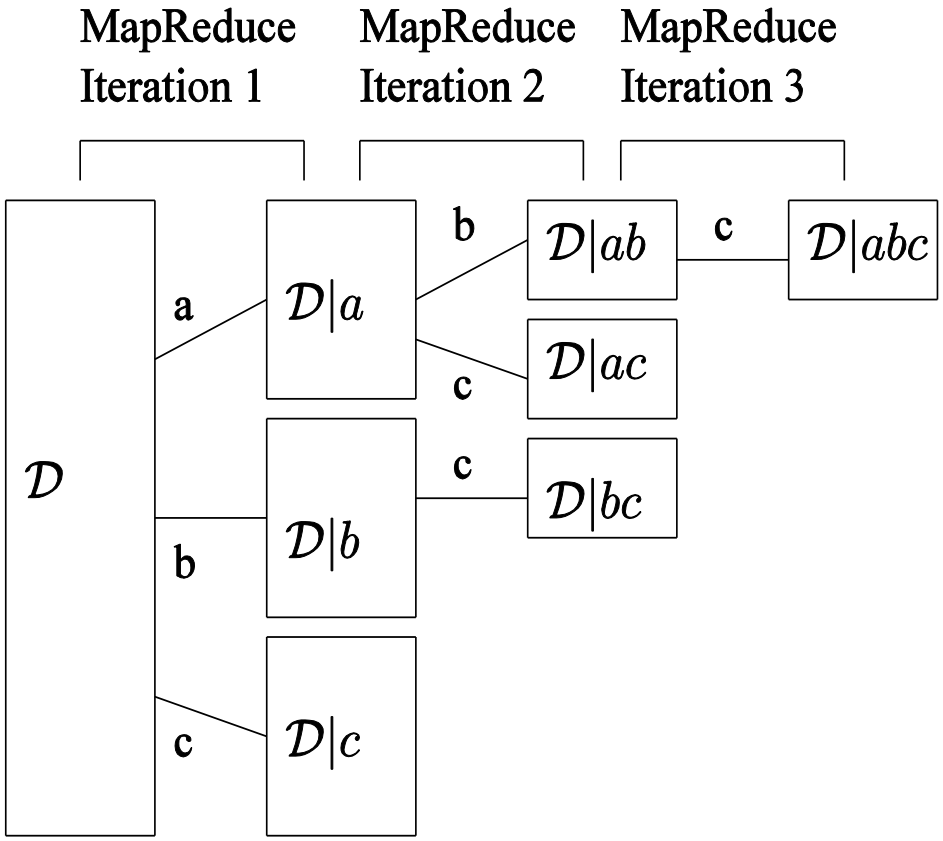
Example of Projection



Example of Projection



Recursive Projections [H. Li, et al. ACM RS 08]



- Recursive projection form a search tree
- Each node is a CDB
- Using the order of items to prevent duplicated CDBs.
- Each level of breath-first search of the tree can be done by a MapReduce iteration.
- Once a CDB is small enough to fit in memory, we can invoke FP-growth to mine this CDB, and no more growth of the sub-tree.

Projection using MapReduce

Map inputs (transactions) key="": value	Sorted transactions (with infrequent items eliminated)	Map outputs (conditional transactions) key: value
f a c d g i m p	f c a m p	p: f c a m m: f c a a: f c c: f
a b c f l m o	f c a b m	m: f c a b b: f c a a: f c c: f
b f h j o	f b	b: f
b c k s p	c b p	p: c b
a f c e l p m n	f c a m p	b: c p: f c a m m: f c a a: f c c: f

p:{fcam/fcam/cb} p:3, pc:3

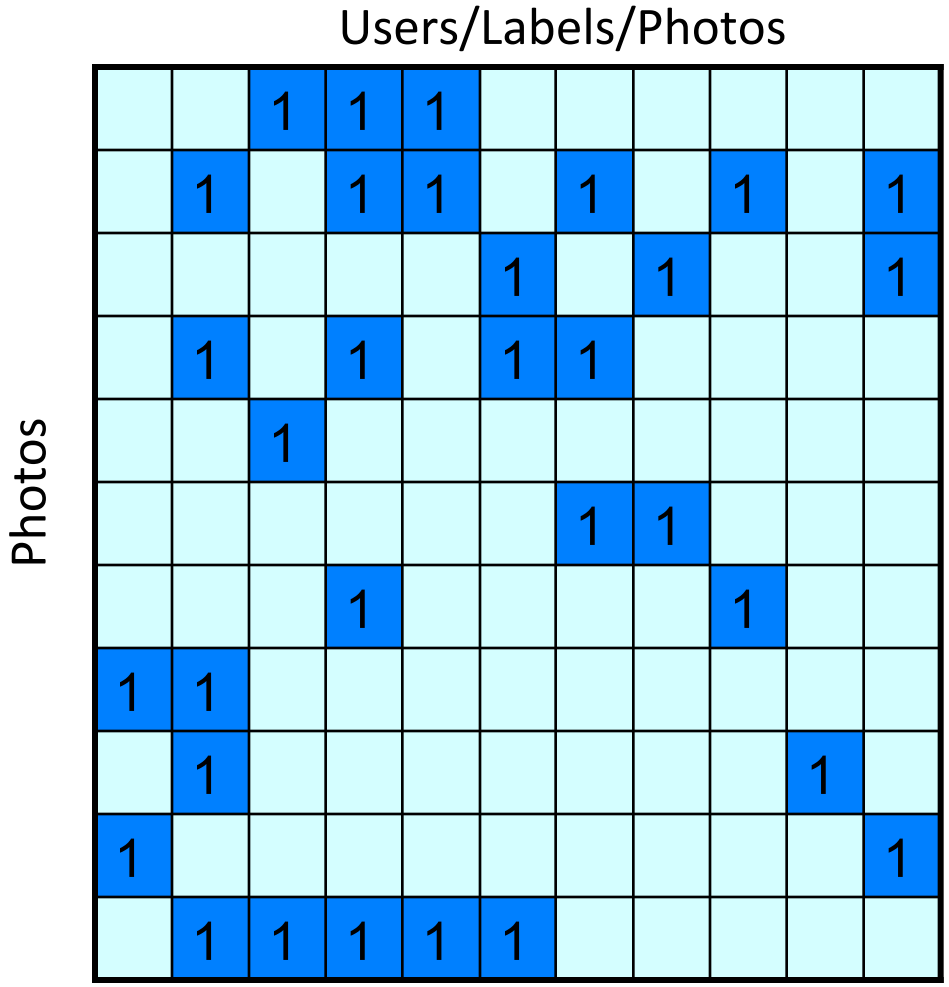
Reduce inputs (conditional databases) key: value	Reduce outputs (patterns and supports) key: value
m: {f c a/f c a/f c a b}	m f:3 m c:3 m a:3 m f c:3 m f a:3 m c a:3 m f c a:3
b: {f c a/f/c}	b:3
a: {f c/f c/f c}	a:3 a f:3 a c:3 a f c:3
c: {f/f/f}	c:3 c f:3

Collaborative Filtering

Based on *membership* so far,
and *memberships* of others

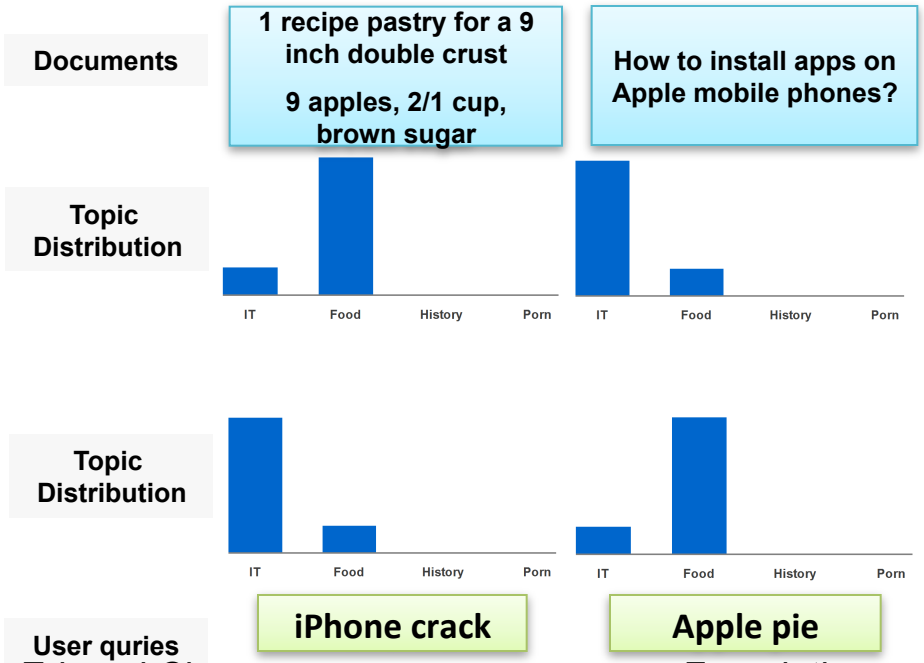
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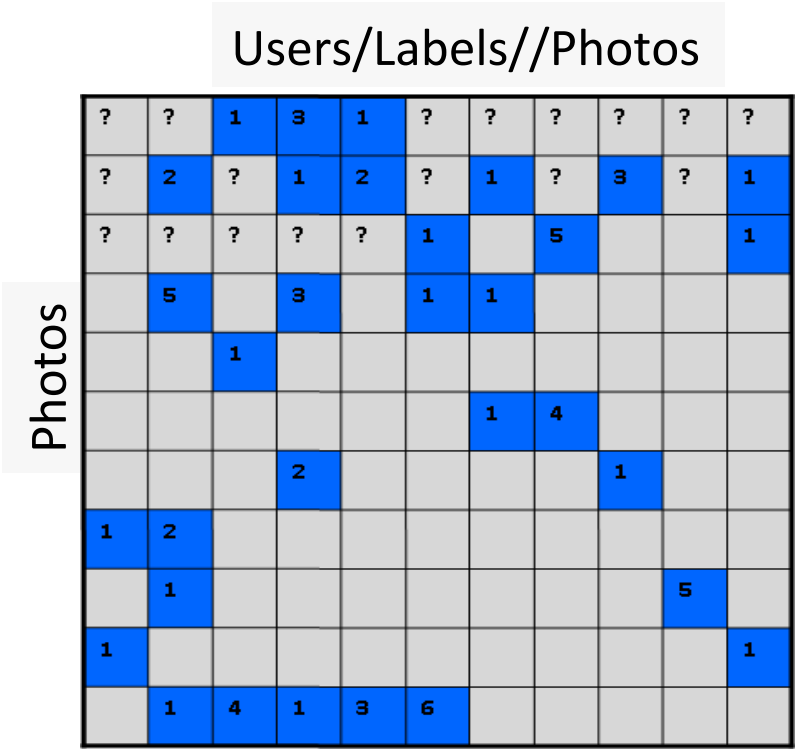
Predict further *membership*



Latent Semantic Analysis

- Search
  - Construct a latent layer for better for semantic matching
- Example:
  - iPhone crack
  - Apple pie





- Collaborative Filtering Apps
  - Recommend Users → Photos
  - Recommend Labels → Photos
  - Recommend Photos → Photos
- Predict the ? In the light-blue cells

Four Shortcomings

- Semantic gap
- Strength of interest
- No effective way to fuse multiple sources
- No consideration of time

Documents, Topics, Words

- A document consists of a number of topics
  - A document is a probabilistic mixture of topics
- Each topic generates a number of words
  - A topic is a distribution over words
  - The probability of the ith word in a document

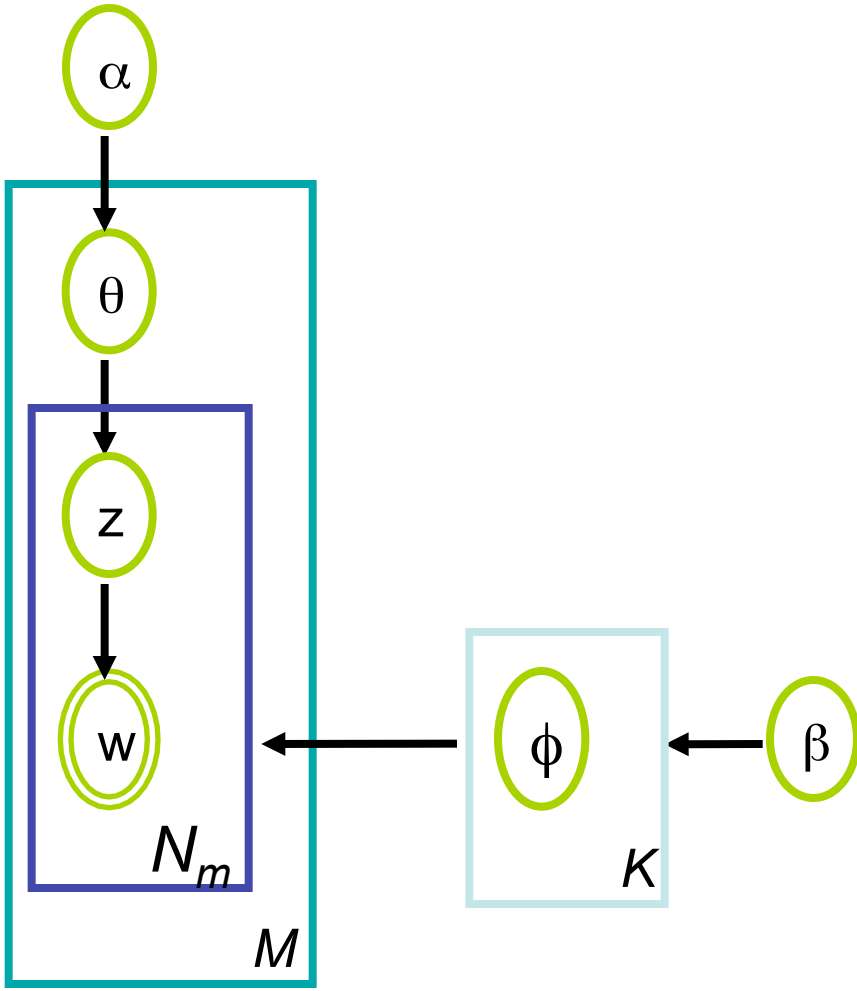
$$P(w_i) = \sum_{j=1}^{T} P(w_i|z_i = j)P(z_i = j)$$

Photos, Topics, Patches

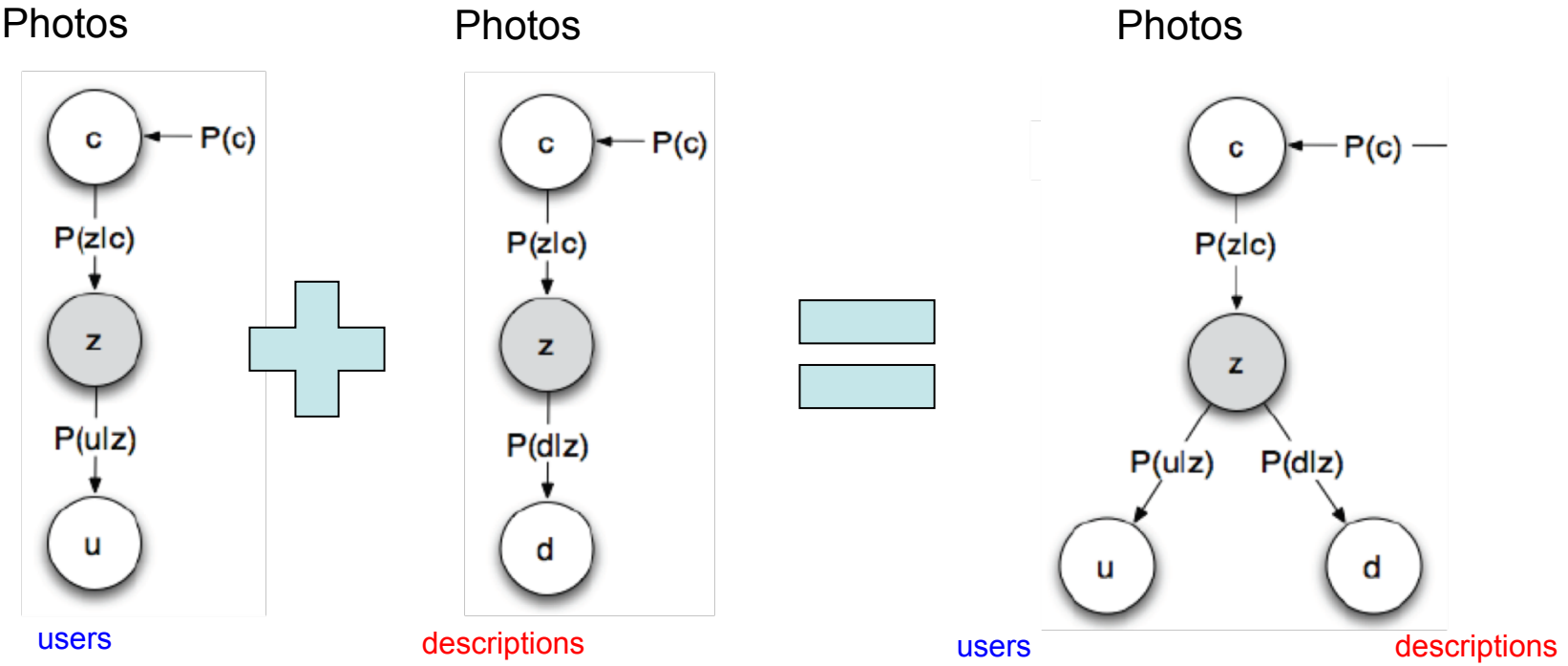
- A photo consists of a number of topics
  - A photo is a probabilistic mixture of topics
  - E.g., where, when, what, and who
- Each topic generates a number of patches
  - A topic is a distribution over patches
  - $$P(w_i) = \sum_{j=1}^{T} P(w_i|z_i = j)P(z_i = j)$$

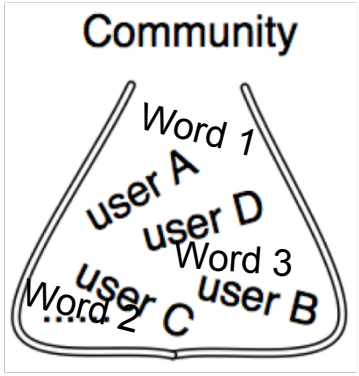
Latent Dirichlet Allocation [M. Jordan 04]

- α: uniform Dirichlet ϕ prior for per document d topic distribution (corpus level parameter)
- β: uniform Dirichlet ϕ prior for per topic z word distribution (corpus level parameter)
- θ_d is the topic distribution of doc d (document level)
- z_{dj} the topic if the jth word in d, w_{dj} the specific word (word level)



Combinational Collaborative Filtering Model (CCF)
【KDD2008】





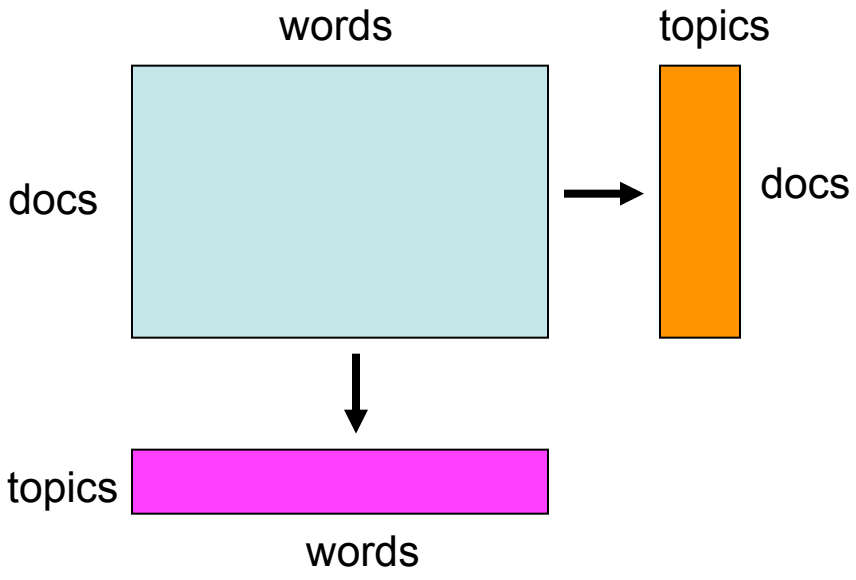
LDA Gibbs Sampling: Inputs And Outputs

Inputs:

1. **Training data**: documents as bags of words
2. Parameter: the number of topics

Outputs:

1. A co-occurrence matrix of topics and documents
2. A co-occurrence matrix of topics and words



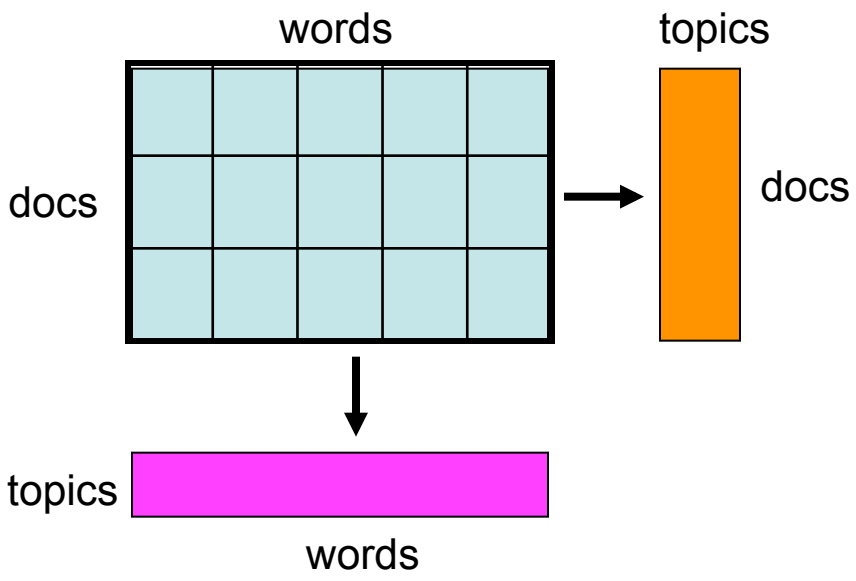
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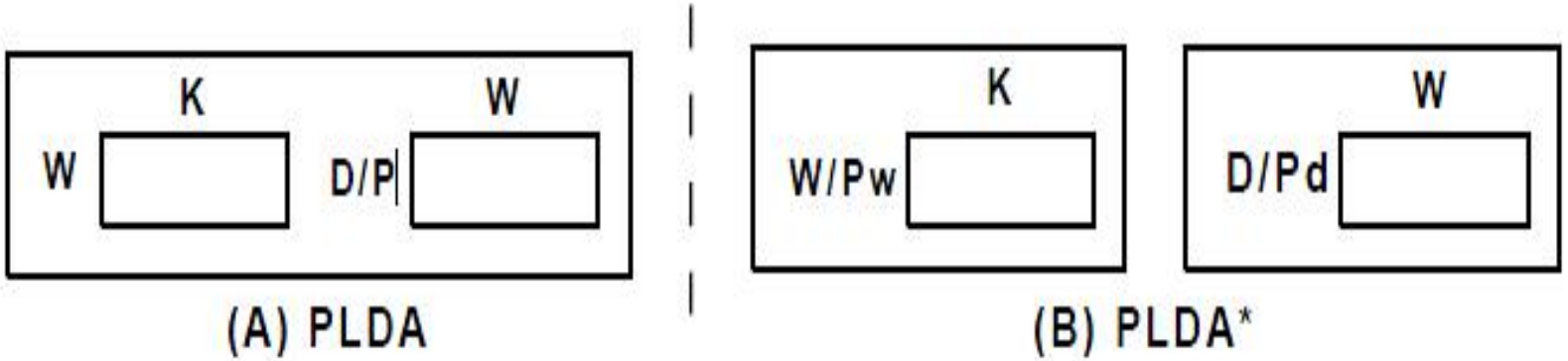
1. A co-occurrence matrix of topics and documents
2. A co-occurrence matrix of topics and words



PLDA+ --- enhanced parallel LDA

[ACM TIST 2010]

- PLDA is restricted by memory: Topic-word matrix has to fit into memory
- Restricted by Amdahl's Law: communication costs too high



PLDA+ --- enhanced parallel LDA

- Take advantage of bag of words modeling: each Pw machine processes vocabulary in a word order
- Pipelining: fetching the updated topic distribution matrix while doing Gibbs sampling

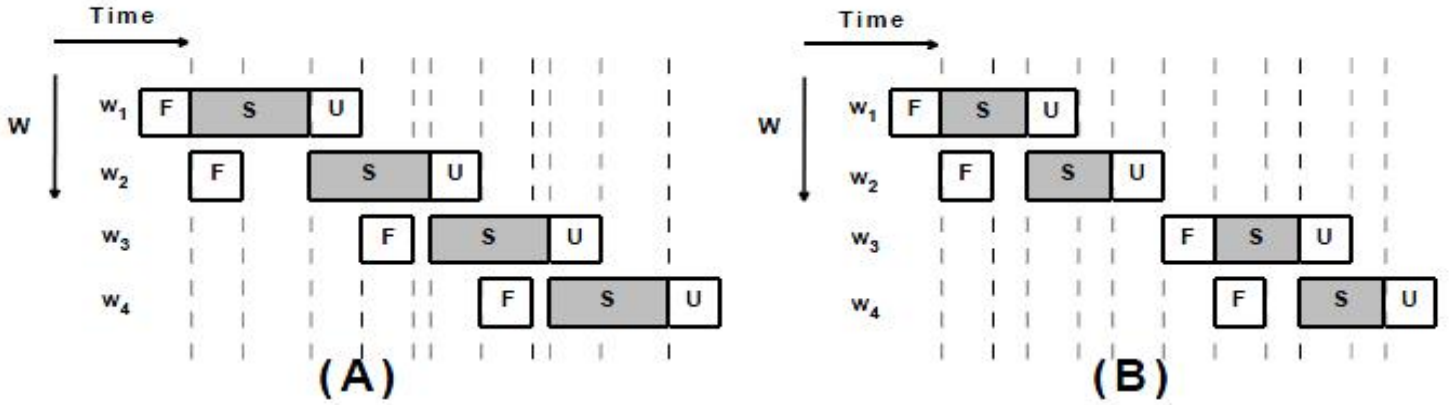
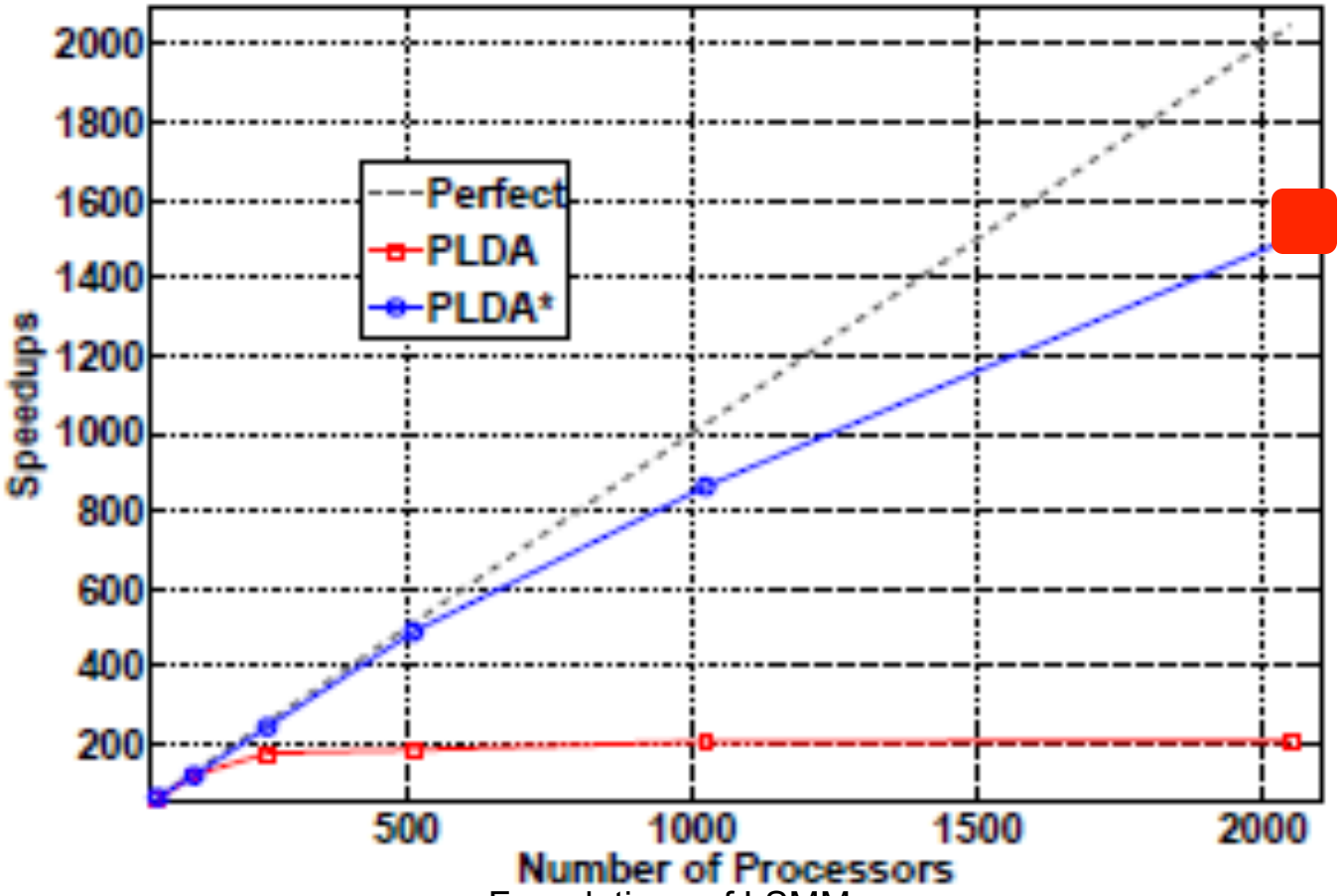


Fig. 4: Pipeline-based Gibbs Sampling in PLDA*. (A): $t_s \geq t_f + t_u$. (B): $t_s < t_f + t_u$.

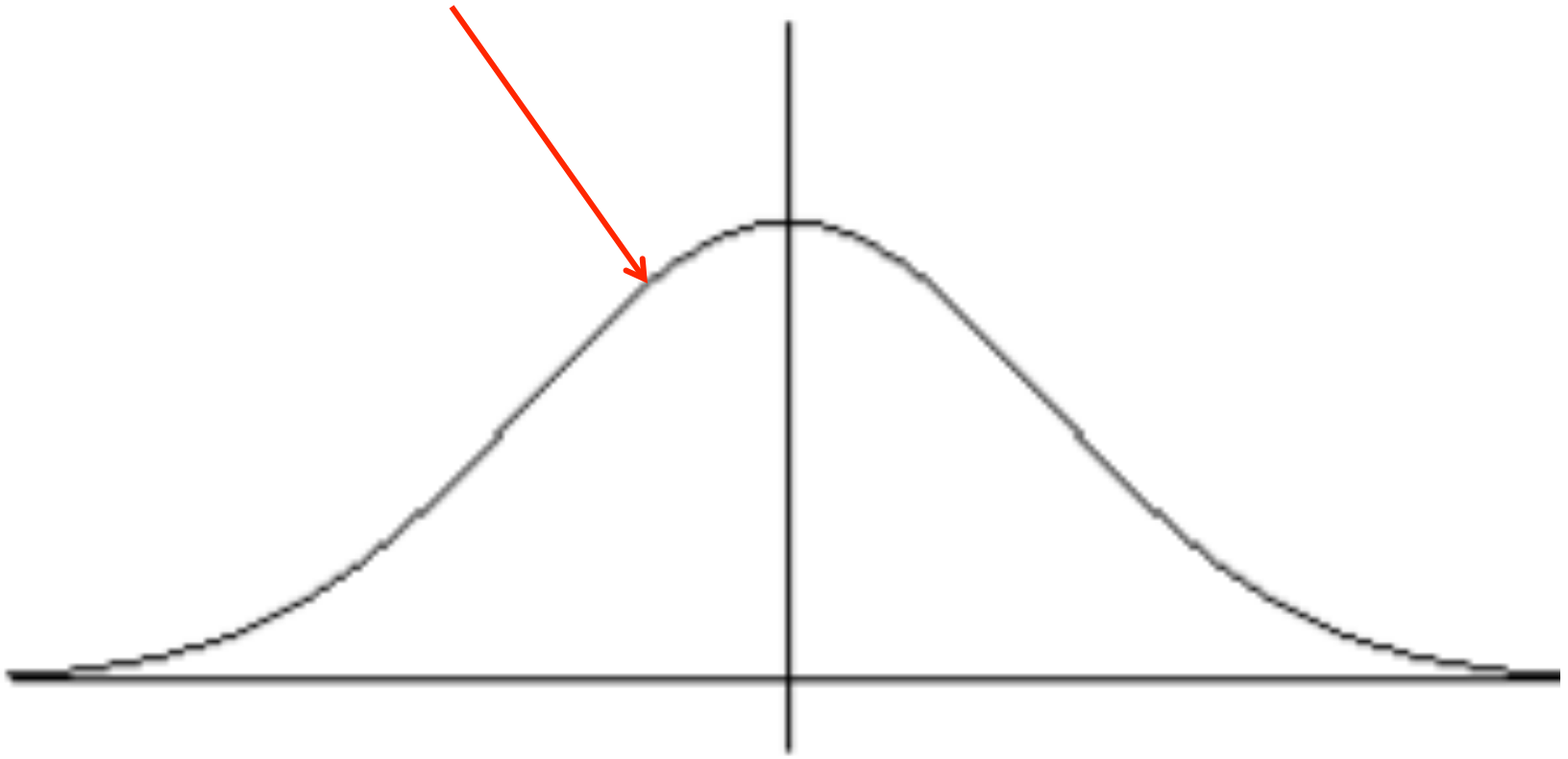
Speedup

1,500x using 2,000 machines



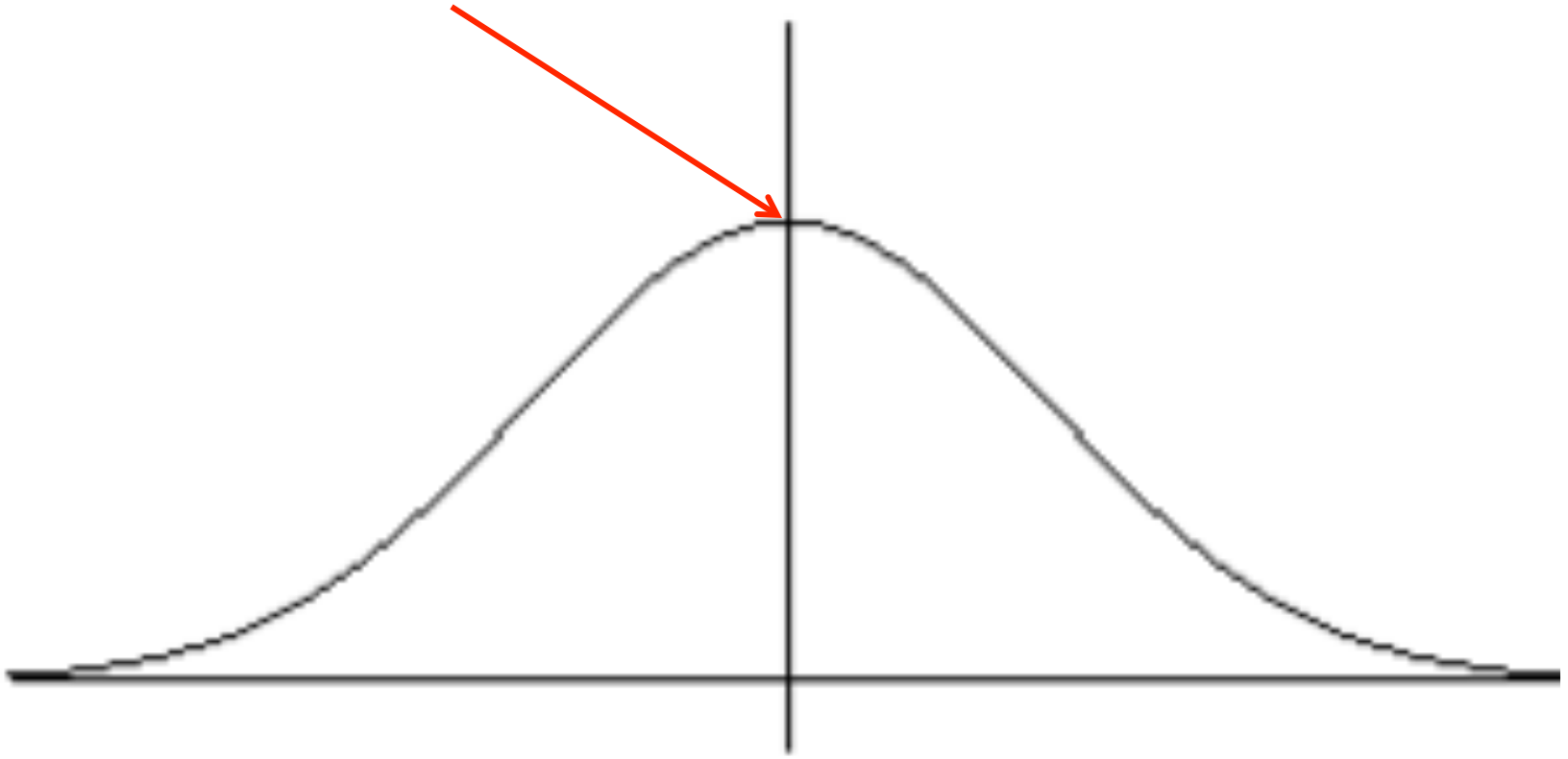
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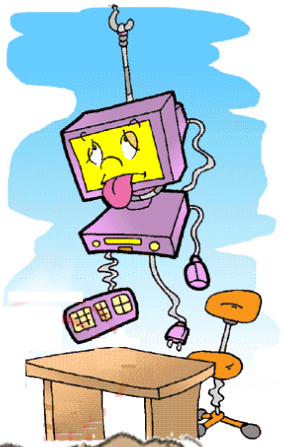
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SVM Bottlenecks

Time consuming – 1M dataset, 8 days





Memory consuming – 1M dataset, 10G







Matrix Factorization Alternatives

Factorization	Cost
QR	$O(\frac{4}{3}n^3)$
LU	$O(\frac{2}{3}n^3)$
Cholesky	$O(\frac{1}{3}n^3 + 2n^2)$
LDL^T	$O(\frac{1}{3}n^3)$
Incomplete Cholesky	$O(p^2n)$
Kronecker	$O(2n^2)$

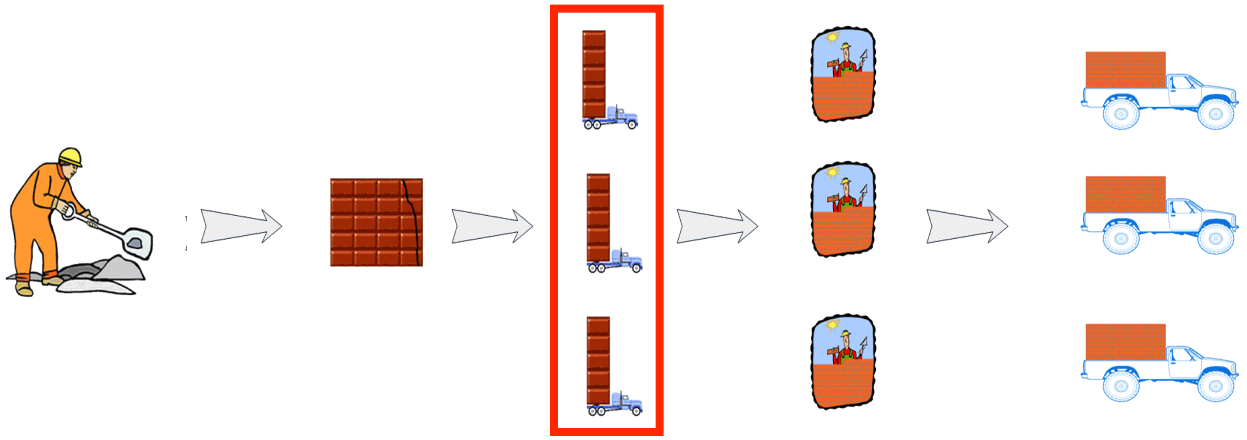
exact

approximate

PSVM [E. Chang, et al, NIPS 07]

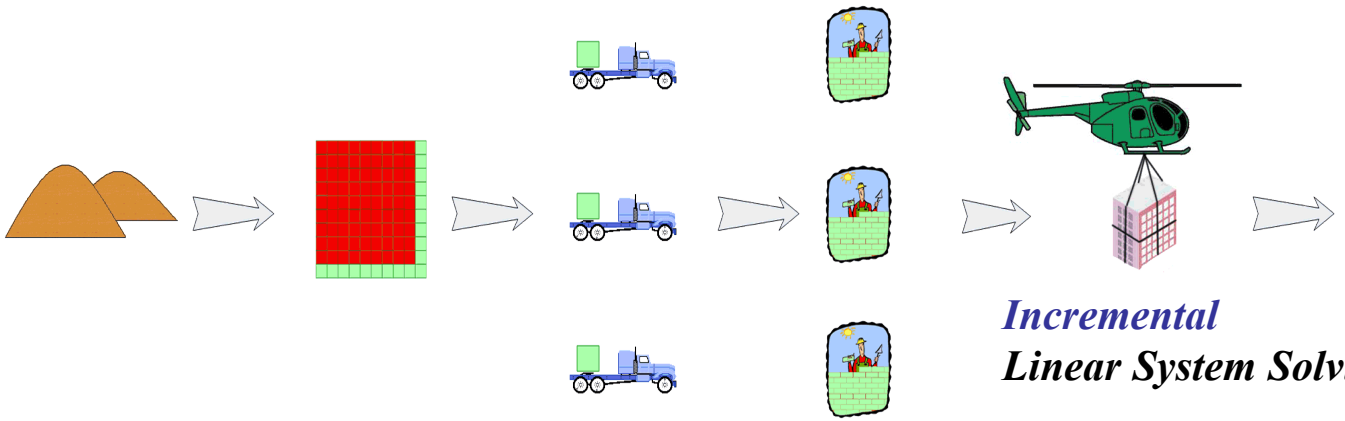
- Column-based ICF
  - Slower than row-based on single machine
  - Parallelizable on multiple machines
- Changing IPM computation order to achieve parallelization

Parallelized and Incremental SVM



Raw Data *Kernel Matrix* *ICF* *Matrix Multiplication*

Matrix Summation



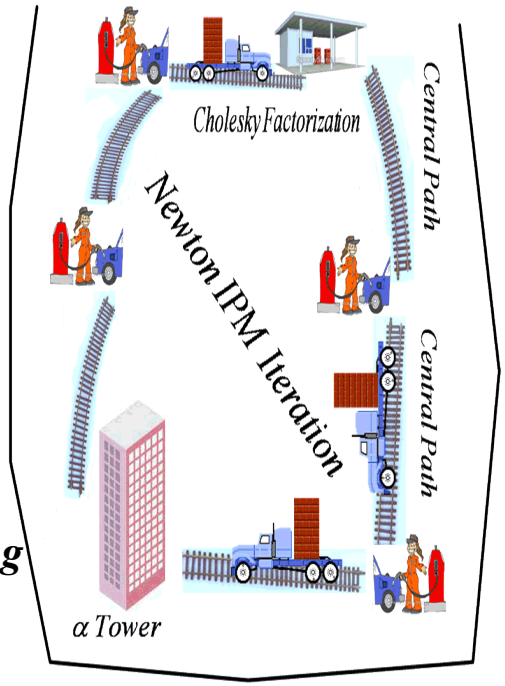
Incremental *Data*

Incremental *Kernel Matrix*

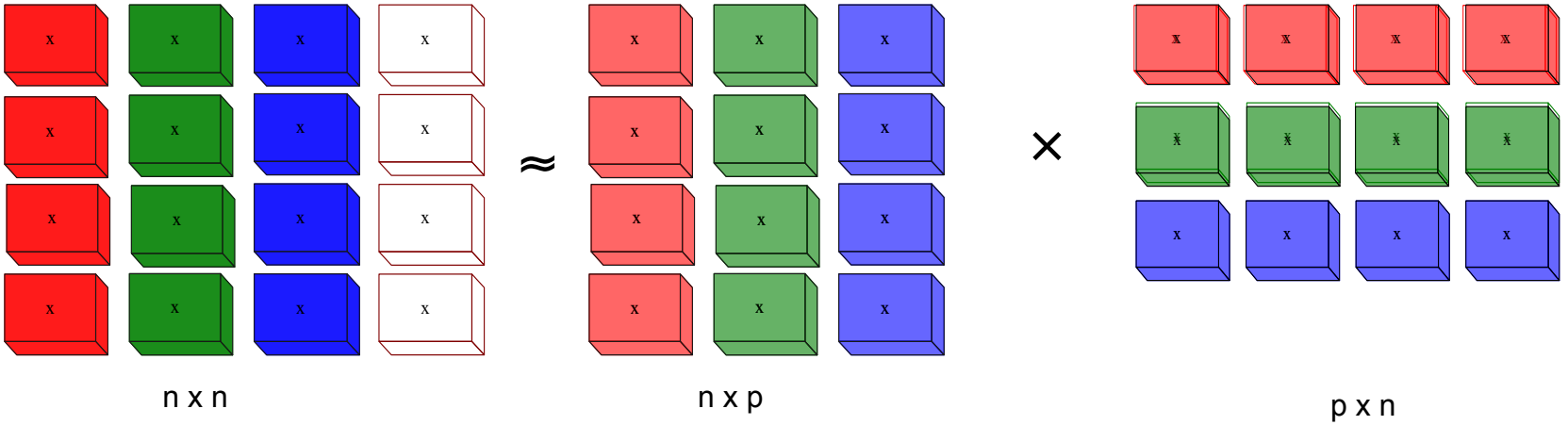
Incremental *ICF*

Incremental *Matrix Multiplication*

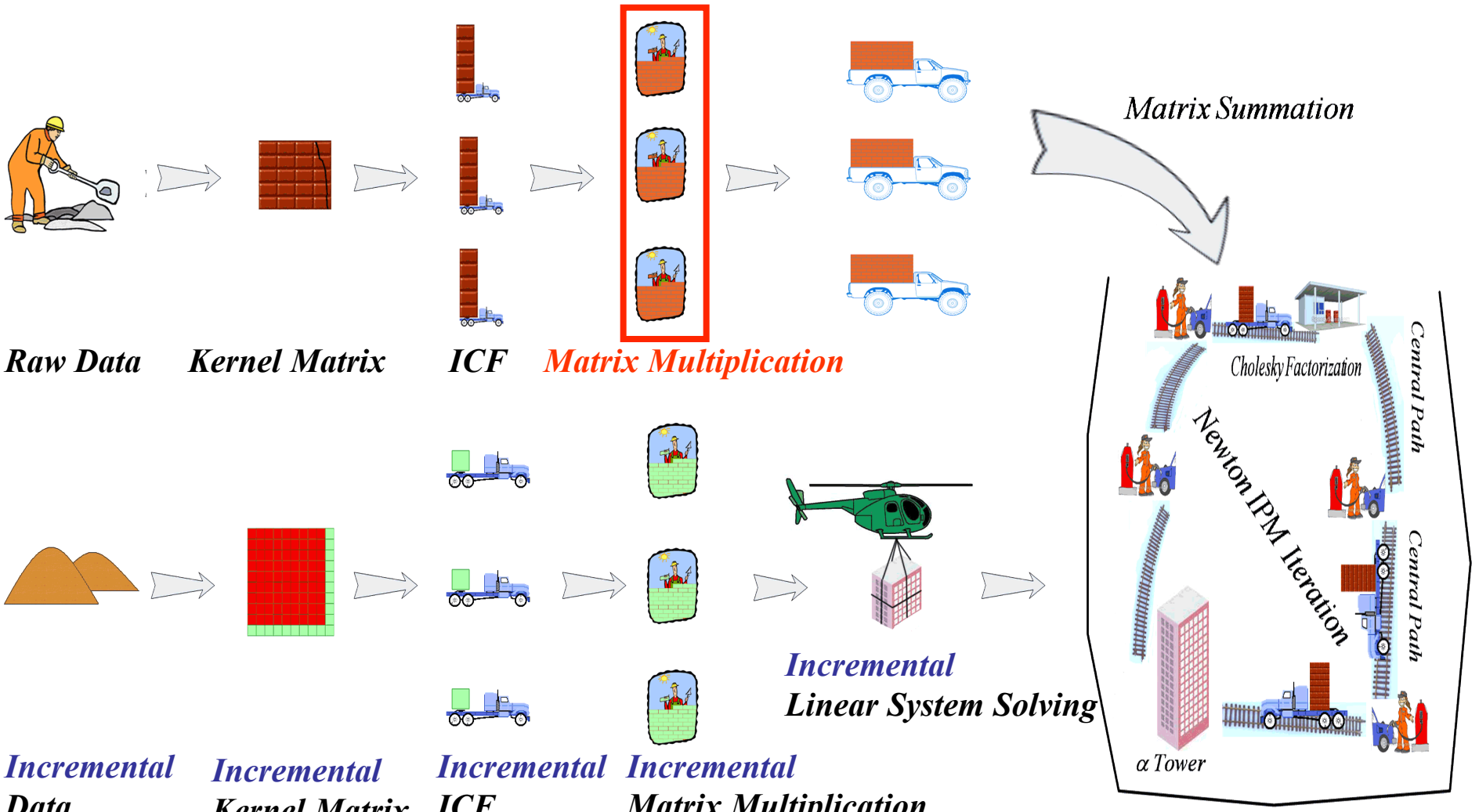
Incremental *Linear System Solving*



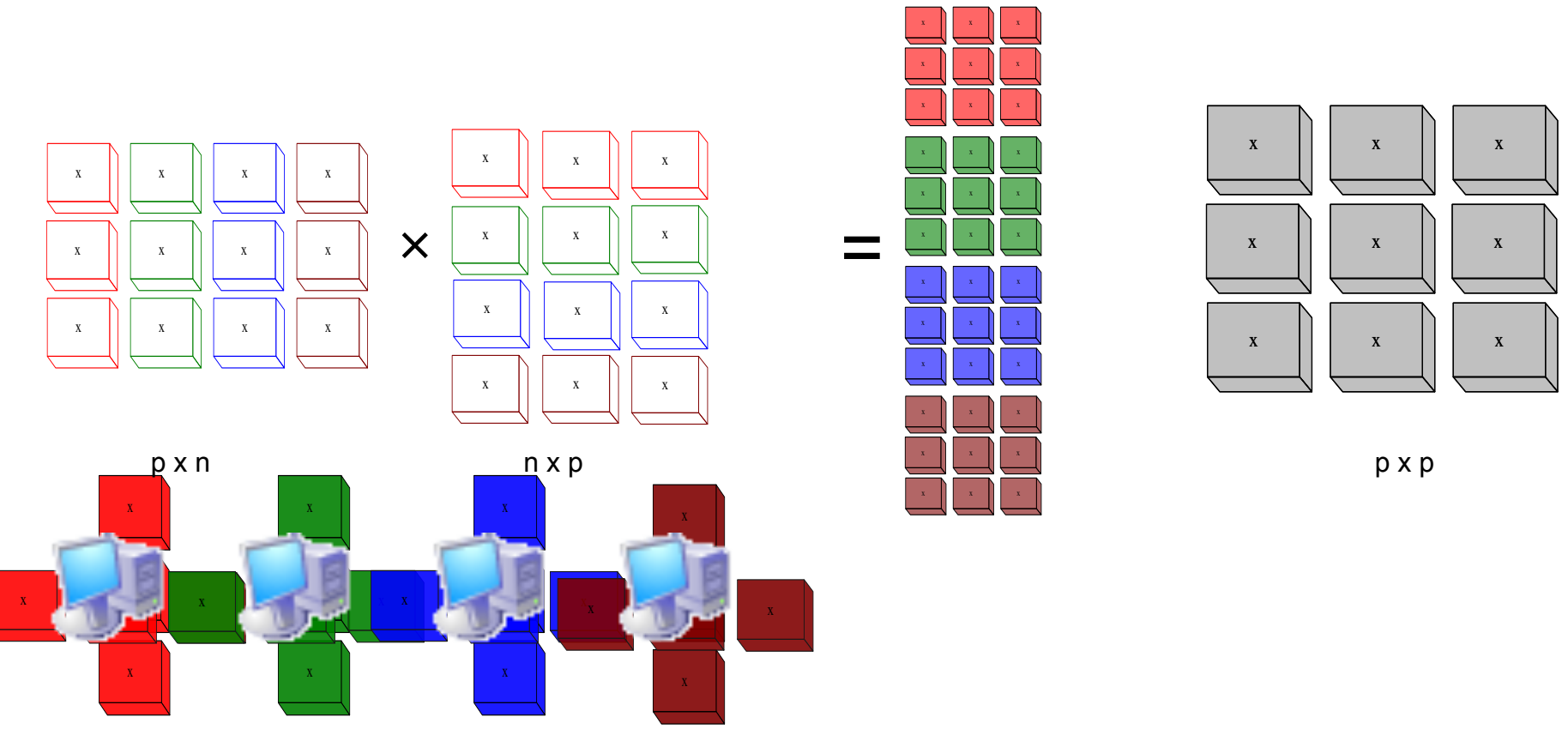
Incomplete Cholesky Factorization (ICF)



Parallelized and Incremental SVM



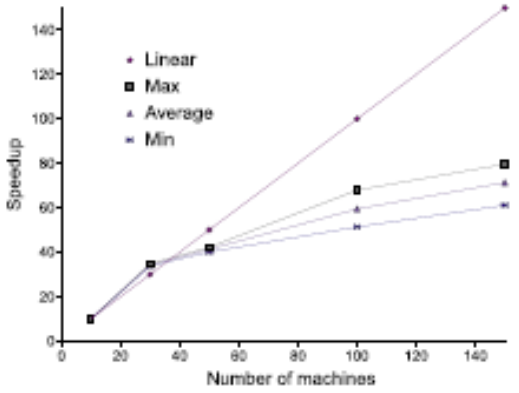
Matrix Product

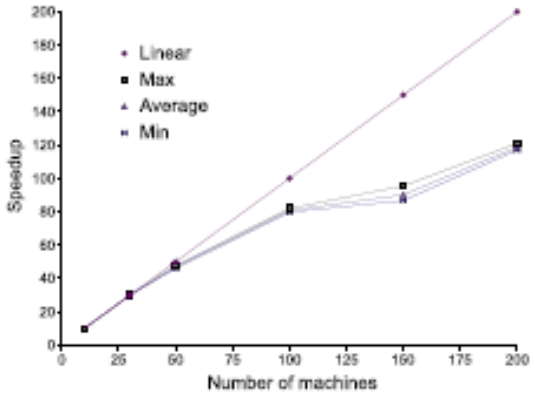


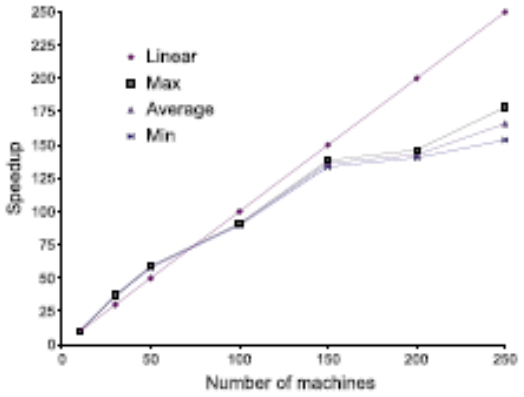
Speedup

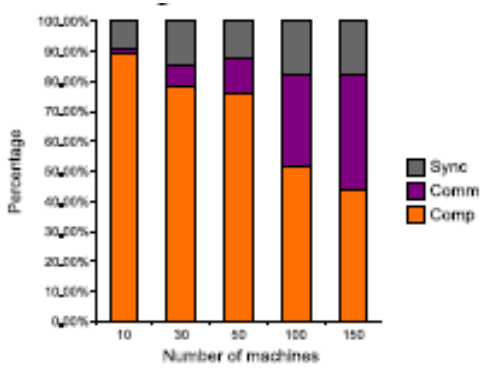
	Image (200k)			CoverType (500k)			RCV (800k)		
Machines	Time (s)		Speedup	Time (s)		Speedup	Time (s)		Speedup
10	1,958	(9)	10*	16,818	(442)	10*	45,135	(1373)	10*
30	572	(8)	34.2	5,591	(10)	30.1	12,289	(98)	36.7
50	473	(14)	41.4	3,598	(60)	46.8	7,695	(92)	58.7
100	330	(47)	59.4	2,082	(29)	80.8	4,992	(34)	90.4
150	274	(40)	71.4	1,865	(93)	90.2	3,313	(59)	136.3
200	294	(41)	66.7	1,416	(24)	118.7	3,163	(69)	142.7
250	397	(78)	49.4	1,405	(115)	119.7	2,719	(203)	166.0
500	814	(123)	24.1	1,655	(34)	101.6	2,671	(193)	169.0
LIBSVM	4,334	NA	NA	28,149	NA	NA	184,199	NA	NA

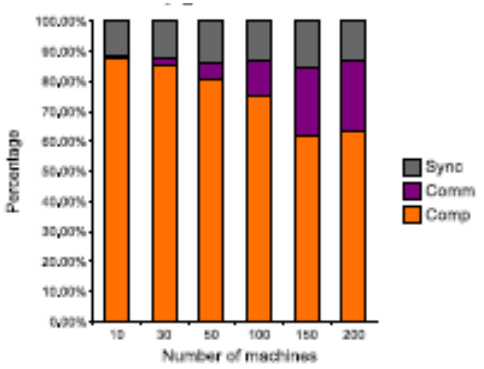
Overheads

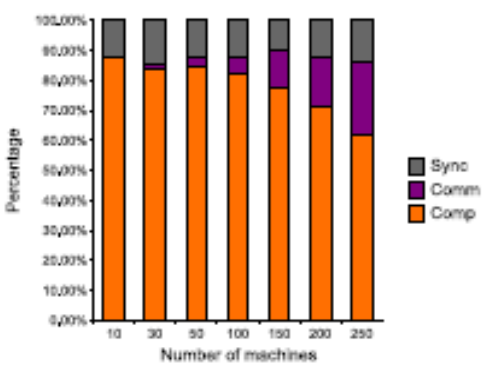

Linear
Max
Average
Min
Speedup
Number of machines


Linear
Max
Average
Min
Speedup
Number of machines


Linear
Max
Average
Min
Speedup
Number of machines


Sync
Comm
Comp
Percentage
Number of machines


Sync
Comm
Comp
Percentage
Number of machines

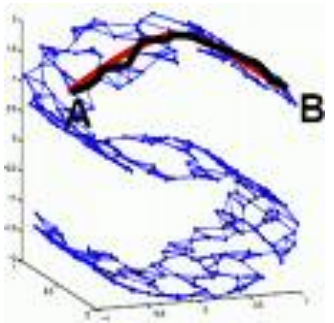

Sync
Comm
Comp
Percentage
Number of machines

Outline

- Motivating Applications
  - Image Annotation
  - Social Media
- Key Subroutines
  - Frequent Itemset Mining [ACM RS 08]
  - Latent Dirichlet Allocation [WWW 09, TIST 10]
  - Support Vector Machines [NIPS 07]
  - Clustering [ECML 08, PAMI 10]
  - UserRank [WWW 10, VLDB 10]
- Distributed Computing Perspectives

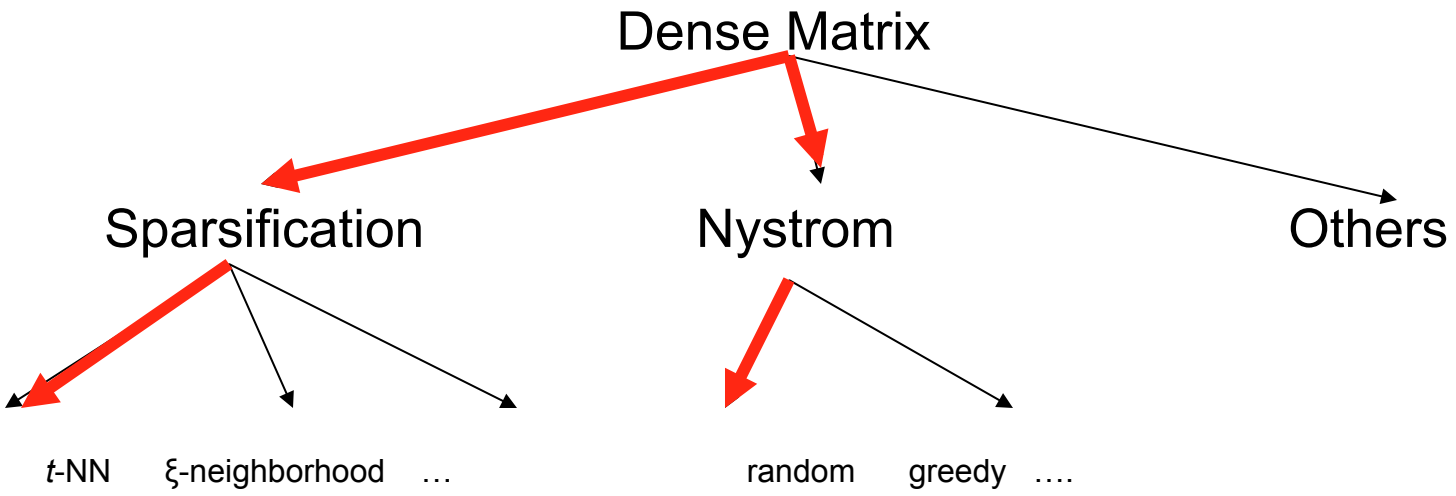
Spectral Clustering [A. Ng, M. Jordan]

- Important subroutine in tasks of machine learning and data mining
  - Exploit *pairwise similarity* of data instances
  - More effective than traditional methods e.g., k-means
- Key steps
  - Construct pairwise similarity matrix
    - e.g., using Geodisc distance
  - Compute the Laplacian matrix
  - Apply eigendecomposition
  - Perform *k*-means



Scalability Problem

- Quadratic computation of nxn matrix
- Approximation methods



Sparsification vs. Sampling

- Construct the dense similarity matrix S
- Sparsify S
- Compute Laplacian matrix L

$$L = I - D^{-1/2} S D^{-1/2}, \quad D_{ii} = \sum_{j=1}^{n} S_{ij}$$

- Apply *ARPACLK* on L
- Use *k*-means to cluster rows of V into *k* groups

- Randomly sample *l* points, where *l* << *n*
- Construct dense similarity matrix [A B] between *l* and *n* points
- Normalize A and B to be in Laplacian form

  $R = A + A^{-1/2} B B^T A^{-1/2}$;

  $R = U \Sigma U^T$
- *k*-means

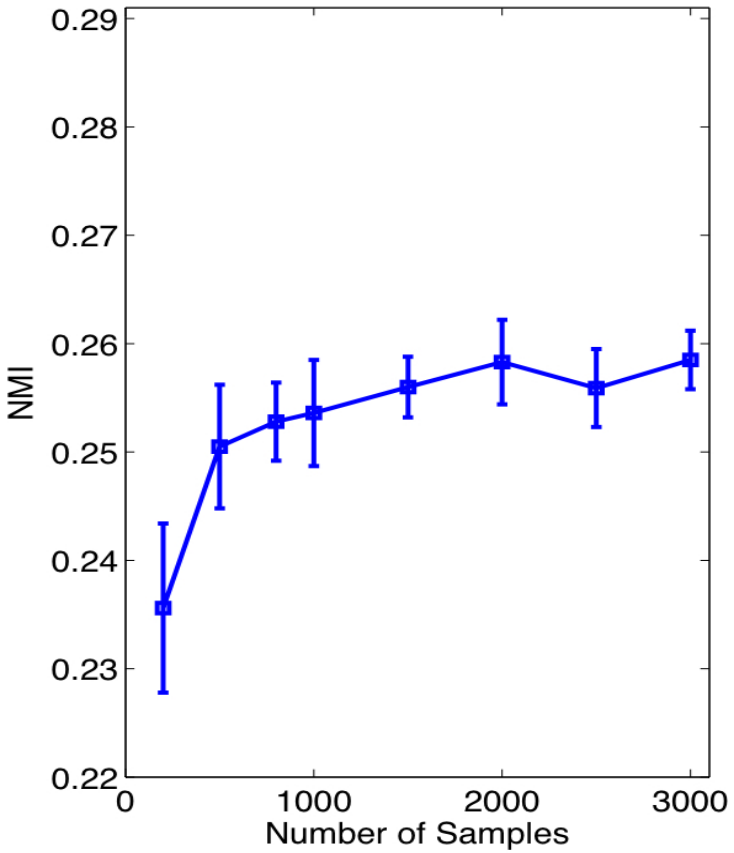
Empirical Study [Song, et al., PAMI 10]

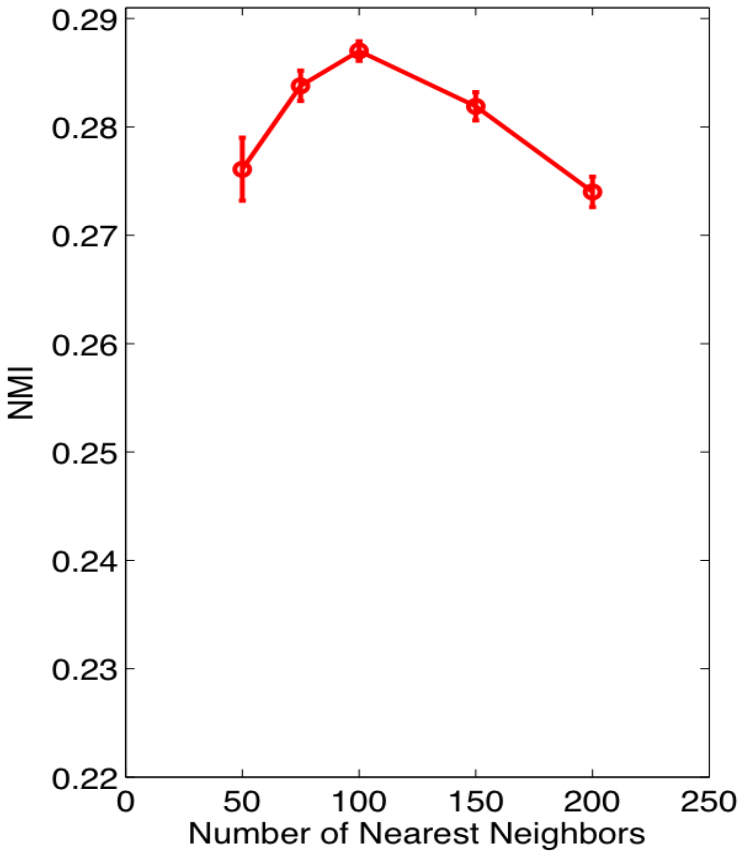
- Dataset: RCV1 (Reuters Corpus Volume I)
  - A filtered collection of *193,944* documents in *103* categories
- Photo set: PicasaWeb
  - *637,137* photos
- Experiments
  - Clustering quality vs. computational time
    - Measure the similarity between CAT and CLS
    - Normalized Mutual Information (NMI)

$$NMI(CAT;CLS) = \frac{I(CAT;CLS)}{\sqrt{H(CAT)H(CLS)}}$$

  - Scalability

NMI Comparison (on RCV1)





Nystrom method

Sparse matrix approximation

Speedup Test on *637,137* Photos

- K = 1000 clusters

	Eigensolver		k-means	
Machines	Time (sec.)	Speedup	Time (sec.)	Speedup
1	—	—	—	—
2	8.074×10^4	2.00	3.609×10^4	2.00
4	4.427×10^4	3.65	1.806×10^4	4.00
8	2.184×10^4	7.39	8.469×10^3	8.52
16	9.867×10^3	16.37	4.620×10^3	15.62
32	4.886×10^3	33.05	2.021×10^3	35.72
64	4.067×10^3	39.71	1.433×10^3	50.37
128	3.471×10^3	46.52	1.090×10^3	66.22
256	4.021×10^3	40.16	1.077×10^3	67.02

- Achiever linear speedup when using 32 machines, after that, sub-linear speedup because of increasing communication and sync time

Sparsification vs. Sampling

	Sparsification	Nystrom, random sampling
Information	Full n x n similarity scores	None
Pre-processing Complexity (bottleneck)	$O(n^2)$ worst case; easily parallizable	O(nl), l << n
Effectiveness	Good	Not bad (Jitendra M., PAMI)

Comparison between Parallel Computing Frameworks

	MapReduce	Project B	MPI
GFS/IO and task rescheduling overhead between iterations	Yes	No +1	No +1
Flexibility of computation model	AllReduce only +0.5	AllReduce only +0.5	Flexible +1
Efficient AllReduce	Yes +1	Yes +1	Yes +1
Recover from faults between iterations	Yes +1	Yes +1	Apps
Recover from faults within each iteration	Yes +1	Yes +1	Apps
Final Score for scalable machine learning	3.5	4.5	5

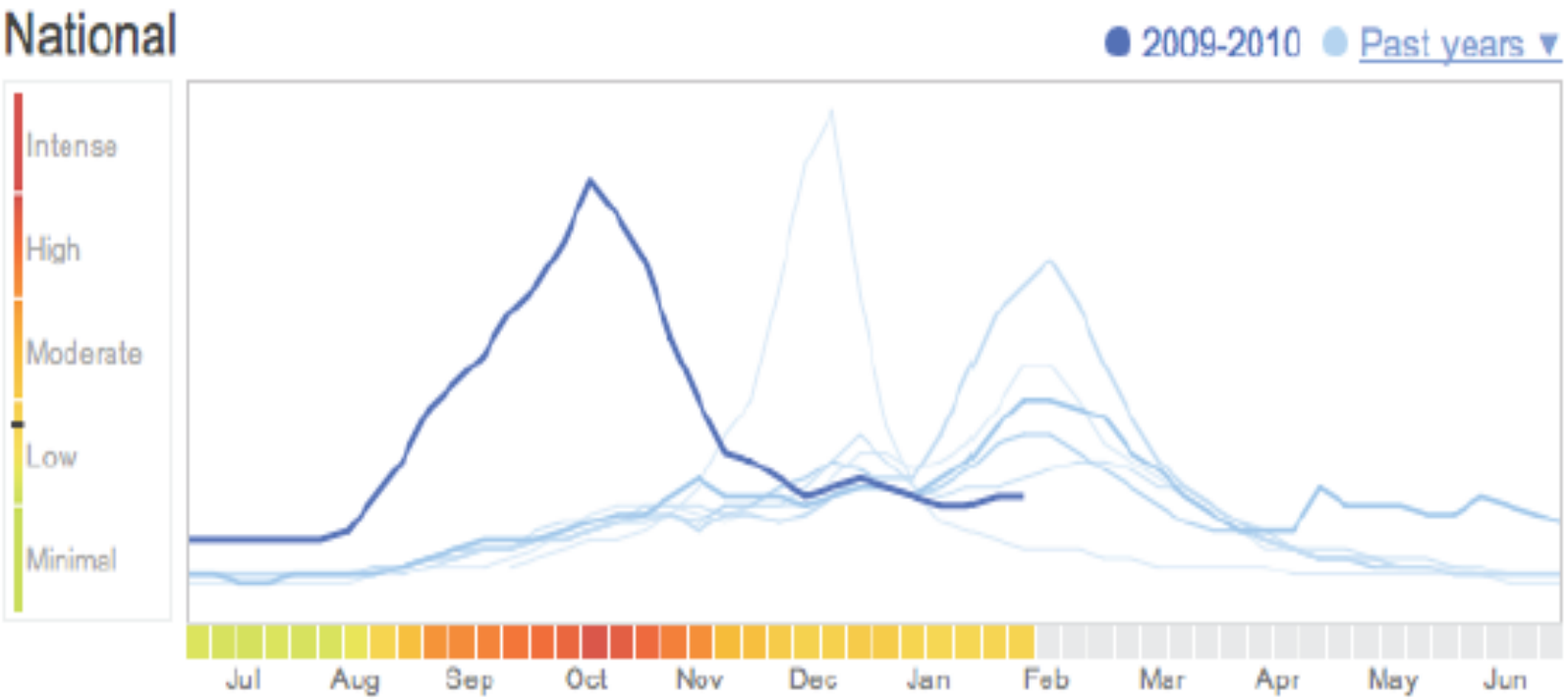
Win in Scale

- Google Translate
- Voice
- Trend Prediction

H1N1 United Nation Report

Explore flu trends - United States

We've found that certain search terms are good indicators of flu activity. Google Flu Trends uses aggregated Google search data to estimate flu activity. Learn more »



Related Publications

- **Large-scale Collaborative Filtering**:
  - PLDA+: Parallel Latent Dirichlet Allocation for Large-Scale Applications, ACM Transactions on Intelligent Systems and Technology (TIST), 2010.
  - Collaborative Filtering for Orkut Communities: Discovery of User Latent Behavior, W.-Y. Chen, J. Chu, Y. Wang, and E. Y. Chang, WWW 2009: 681-690.
  - PFP: Parallel FP-Growth for Query Recommendation, H. Li, Y. Wang, D. Zhang, M. Zhang, and E. Y. Chang, ACM Recommendation Systems, Lausanne, October 2008
  - PVSM --- Parallelizing SVMs on Distributed Machines, E. Y. Chang, et al., NIPS 2007.
  - Parallel Spectral Clustering in Distributed Systems,
    Wen-Yen Chen, Yangqiu Song, Hongjie Bai, Chih-Jen Lin, and E. Y. Chang,
    IEEE Transactions on Pattern Analysis and Machine Intelligence (PAMI), 2010.
- **Social Media**:
  - A Deep Model-based and Data-driven Hybrid Architecture for Image Annotation,
    Zhiyu Wang, Dinying Xia, E. Y. Chang, International Workshop on Very-Large-Scale Multimedia Corpus, Mining and Retrieval, with ACM Multimedia, Florence, October 2010.
  - AdHeat: An Influence-based Diffusion Model for Propagating Hints to Match Ads,
    H.J. Bao and E. Y. Chang, WWW 2010 (best paper candidate), April 2010.
  - Confucius and its Intelligent Disciples, Integrating Search and Social, X. Si, E. Y. Chang, Z. Gyongyi, and M.S. Sun, VLDB, September 2010 .
  - Combinational Collaborative Filtering for Personalized Community Recommendation, W.-Y. Chen, D. Zhang, and E. Y. Chang, KDD 2008: 115-123.

Reading

- Foundations of Large-Scale Multimedia Information Management and Retrieval, E. Y. Chang, Springer, 2011
  - Chapter #10 Indexing
  - Chapter #11 PSVM
  - Chapter #12 PLDA